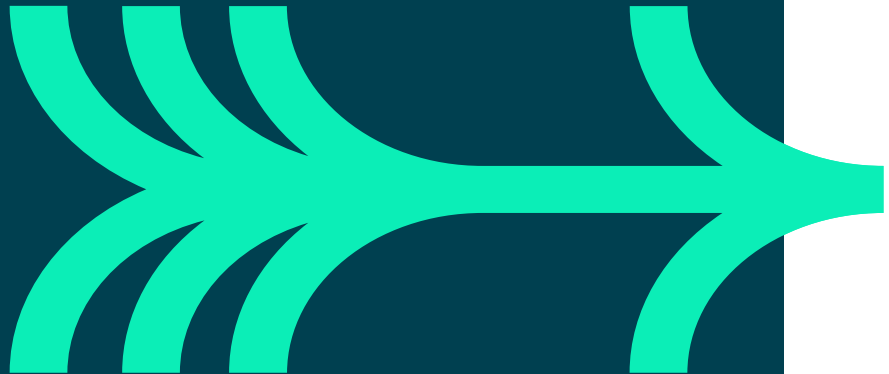




MACHINE LEARNING WITH GCP AND PYTHON - DAY 2



Learning objectives

- Flow Control with Python
- Connecting and ETL Equifax customer data from GCP into python notebooks
- Regression Modelling & Evaluation
- Classification Modelling & Evaluation

Expected prior knowledge

- Have attended Day 1 of this workshop



Flow control

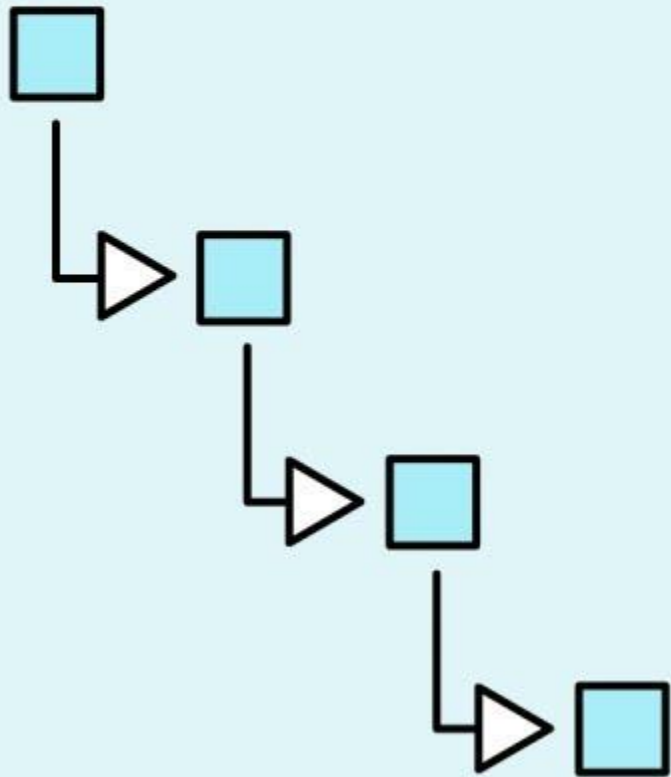


What is control flow?

Control flow controls the order in which we do things:

- **Sequence**
 - Running code step by step, in order.
- **Selection**
 - Deciding which lines of code should run.
- **Iteration / Looping**
 - Doing the same thing many times, i.e., in a loop.

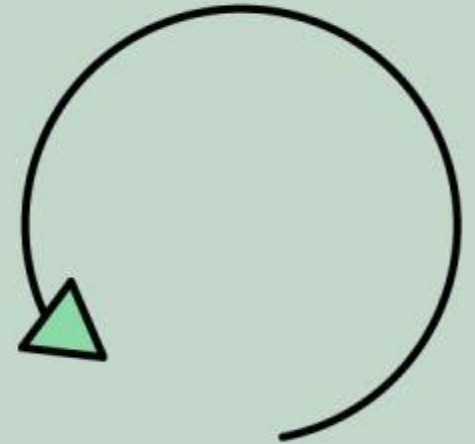
SEQUENCES



SELECTIONS



LOOPS

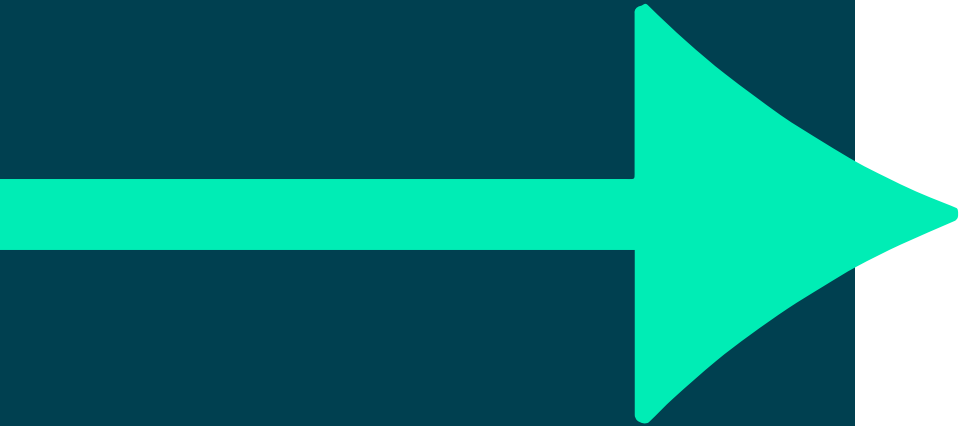




Control flow - selection

Making decisions

- **if:** do this.
- **else:** do something else.

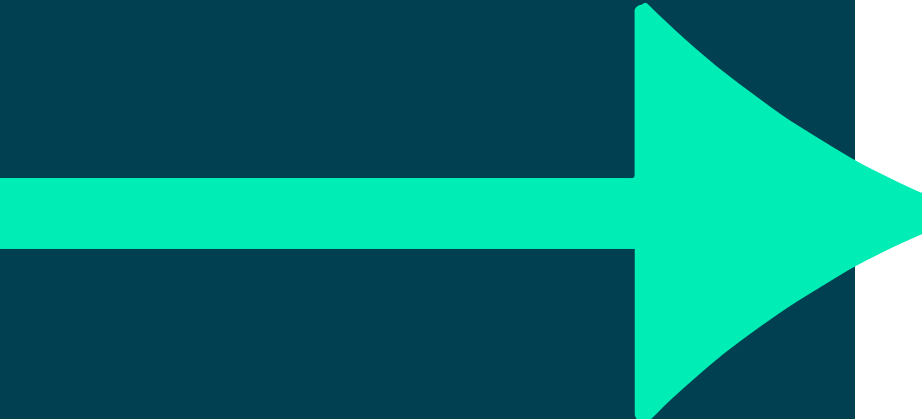




Comparing values

Relational operators for making a selection:

- > Greater than
- >= Greater than or equal to
- < Less than
- <= Less than or equal to
- == Equal to comparison
- != Not equal to comparison



Note!

== is not the same as =



Selection example – IF statement

```
x = 5
```

Condition:
Is x equal to 5?

```
if x == 5:
```

```
↔ print("X equals 5")
```

Yes, it is
... so, do this.



Use the Tab key to include
lines in an IF statement

* In Jupyter Notebooks, it automatically tabs for you.



Selection example – IF ... ELSE

X = 5

Is x equal to 5?

```
if x == 5:  
    print("X equals 5")  
else:  
    print("X does not equal 5")
```

Yes, it is... so, do this.

No, it isn't
... so, do this
instead.



Control flow using 'else if' - elif

```
salary = 2500

if salary > 100000:
    print('Band A')
elif salary > 55000:
    print('Band B')
elif salary > 32000:
    print('Band C')
elif salary > 25000:
    print('Band D')
else:
    print('Band E')
```

Creates a chain of tests.

When a test passes, the other tests will not execute, and the statement ends.

The last **else** is optional.

The **if-elif** conditions must not overlap. Otherwise, the result will depend on their order, which is not a good practice.



Checking for existence

Use the **in** command to check if an item is present in a list.

```
adminIDs = [12,33,84,45,67,36,16,66,67,99]
id = int(input('Enter your ID '))
if id in adminIDs:
    print('Welcome!')
```

Numeric

```
names = ['David','John','Joanne','Sean','Sonia']
if 'Sean' in names:
    print('Sean is in the list!')
```

Strings

```
names = "David,John,Joanne,Sean,Sonia"
if 'John' in names:
    print('John is in the list!')
```

List of characters
(string)



Iterating through a list - For Loop

```
names = ["Bob", "Steve", "Helen"]
```

```
for name in names:  
    print(name)
```

```
Bob  
Steve  
Helen
```

```
Press any key to continue ...
```



Iterating through a list – While Loop

```
names = ["Bob", "Steve", "Helen"]
```

```
i=0      # used as index
```

```
while(i < len(names) ):  
    print(names[i])  
    i += 1
```

```
Bob  
Steve  
Helen
```

```
Press any key to continue ...
```

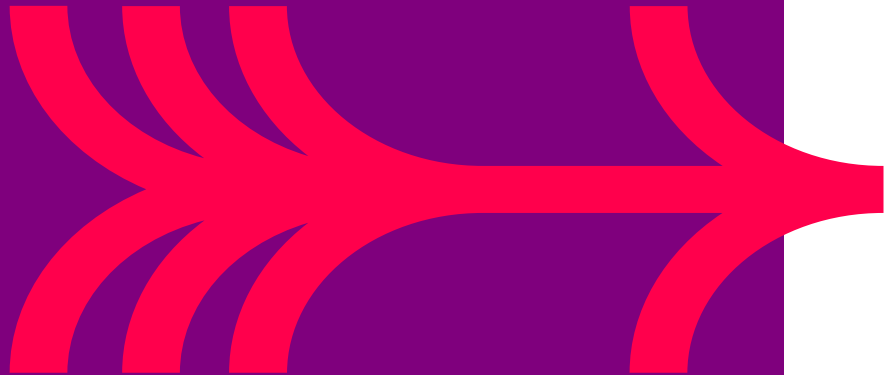


Practicing Flow Control

You will find **Day 2** resources in the Gdrive:
<https://tinyurl.com/mpfjprut>

Open the doc ***Flow Control Exercises.docx*** and
attempt the exercises

You can find the ***Flow Control Solutions.docx*** in
the Solutions folder

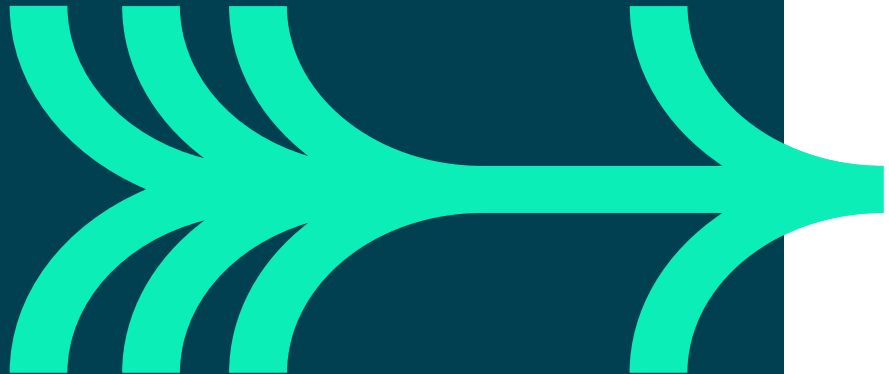




SUPERVISED LEARNING: REGRESSION

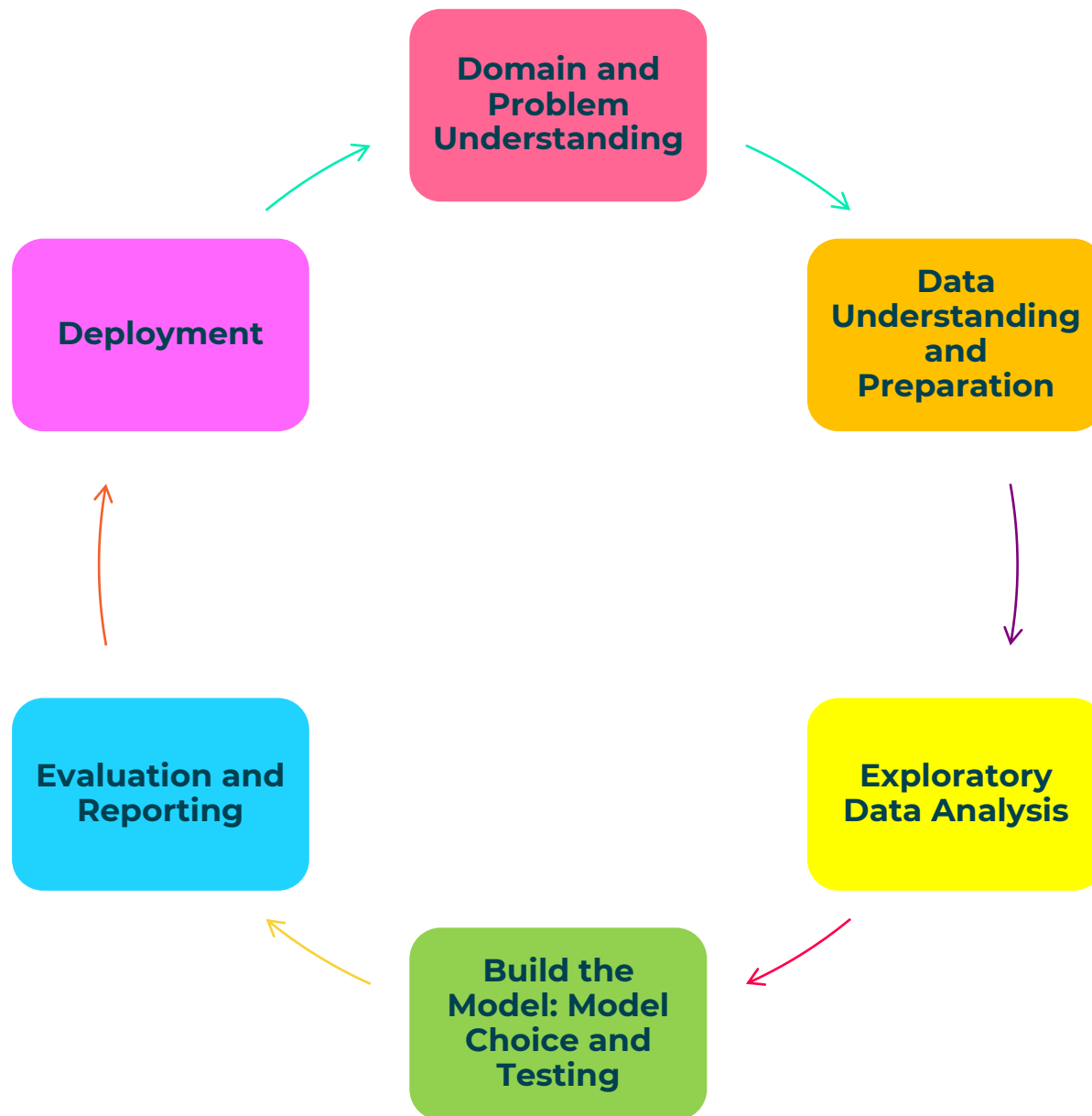
Learning objectives

- Describe regression in the context of machine learning.
- Build simple and multiple linear regression models.
- Understand non-linear regression approaches.
- Evaluate and compare regression models.

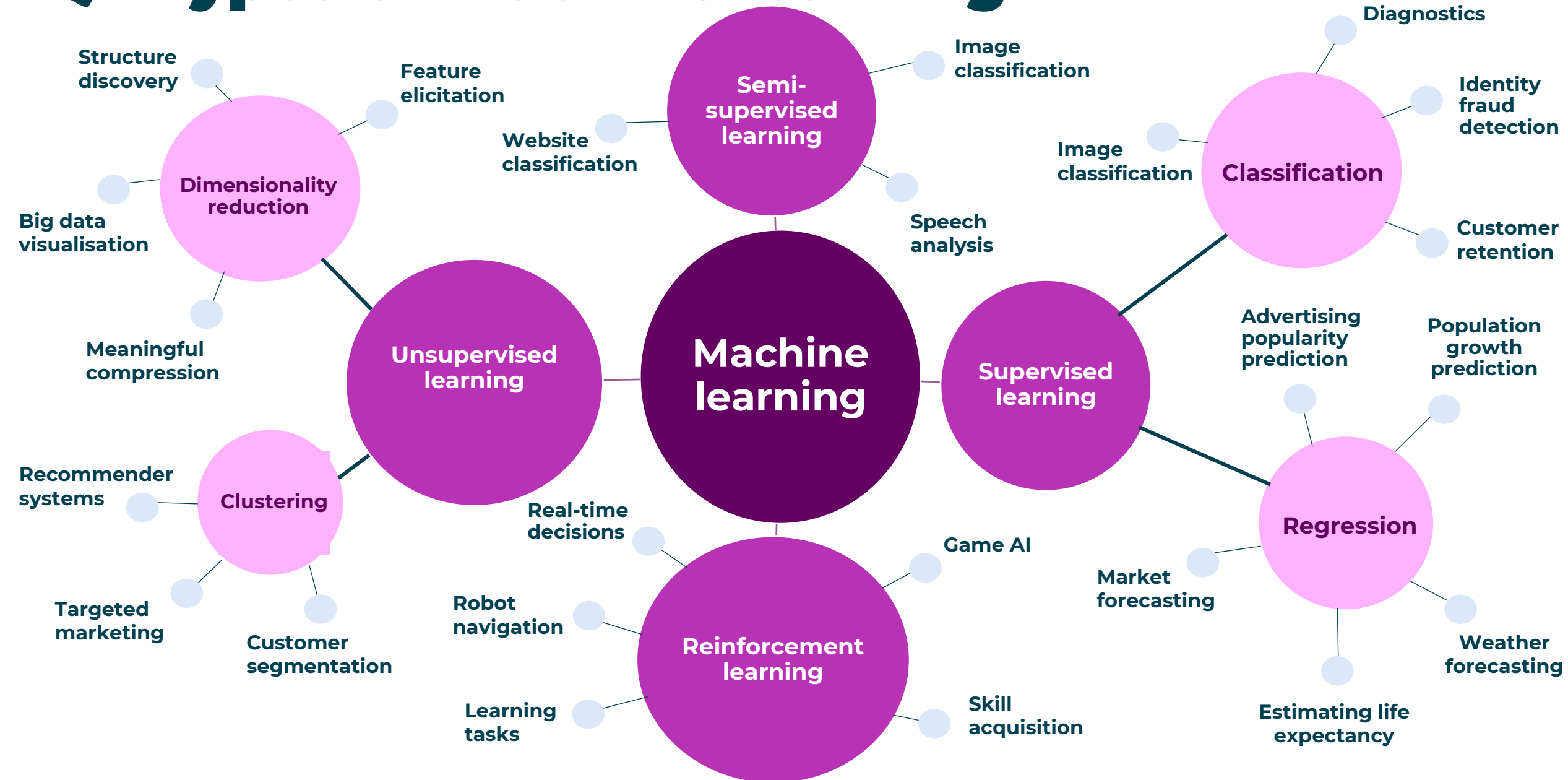


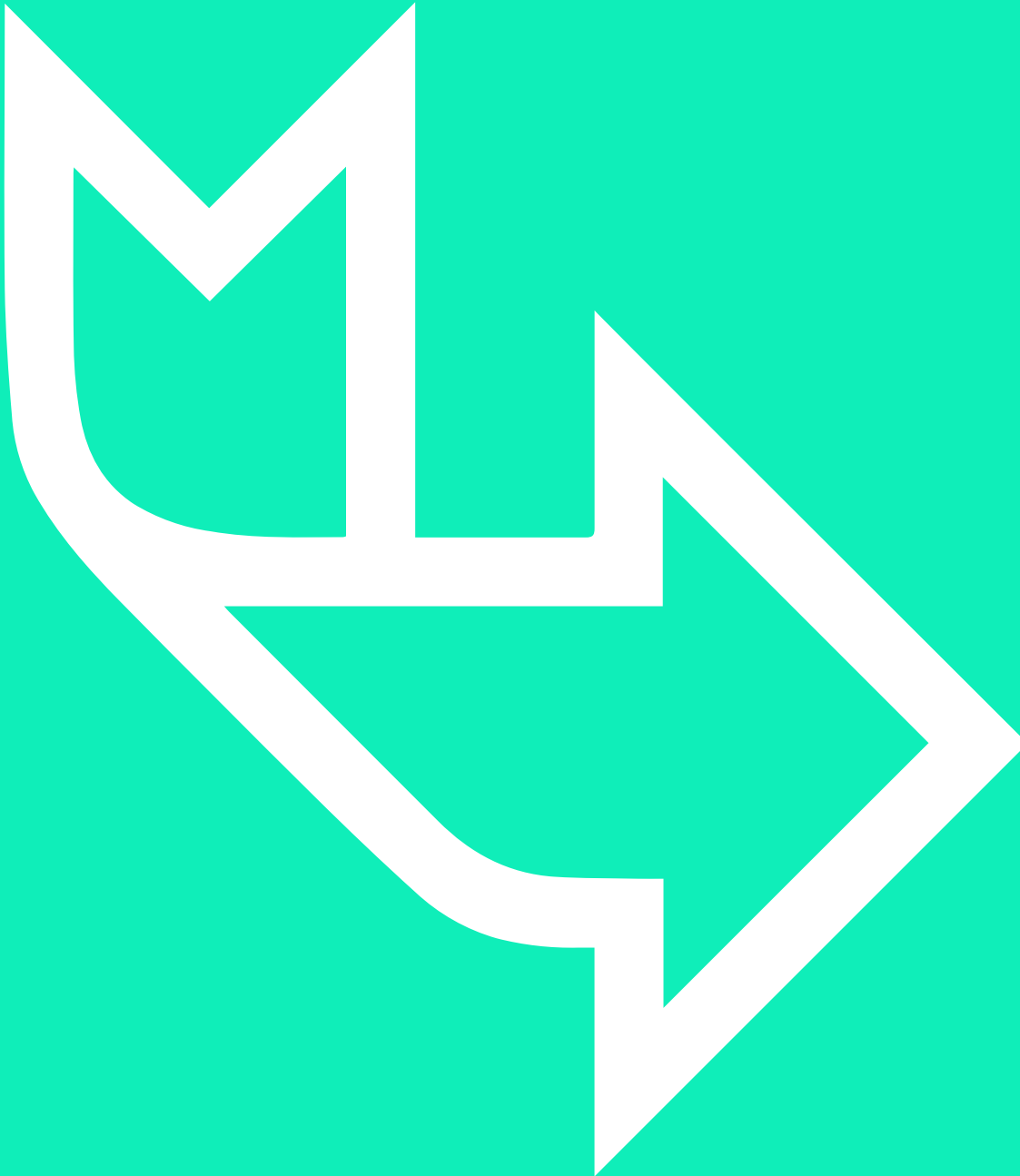


DATA SCIENCE PROJECT LIFECYCLE



QA Types of machine learning



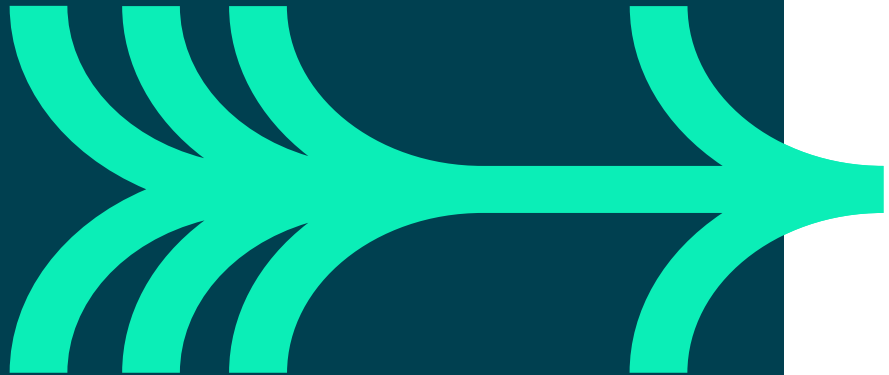


Assignment: ETL from GCP into Python - Pandas

- 1) Download:
Training_Equifax_day_2_GCP_ETL
Python notebook
- 2) Upload notebook to google colab within
your gcpq.qa@gmail.com account
- 3) Attempt Tasks described within
notebook. You are finished when the data
is locally cached / saved



SUPERVISED LEARNING: REGRESSION



Learning objectives

- Describe regression in the context of machine learning.
- Build simple and multiple linear regression models.
- Understand non-linear regression approaches.
- Evaluate and compare regression models.

Expected prior knowledge

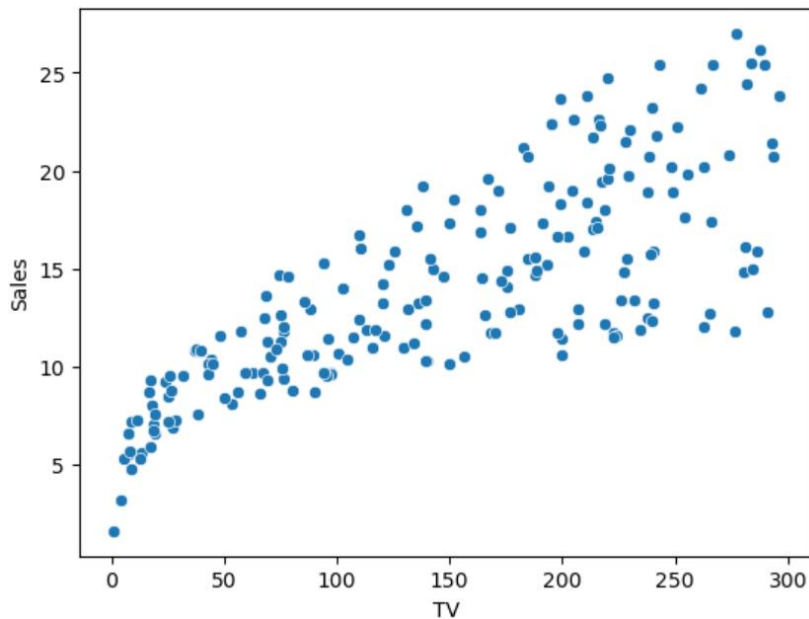
- Nothing is assumed about your background.



Regression

THE GOAL

- Regression is used for **predicting a continuous variable**, y .
- We ask the question: what variable might it be useful to predict?
- This variable can be placed on the vertical axis of a scatter plot, and we can try different variables on the x axis to work towards predicting y accurately.
- Example: Predicting product sales from TV advertising:





POSSIBLE MODELS

- Linear Regression
- Polynomial Regression
- Splines
- Decision Tree
- Random Forest
- XGBoost

And more!



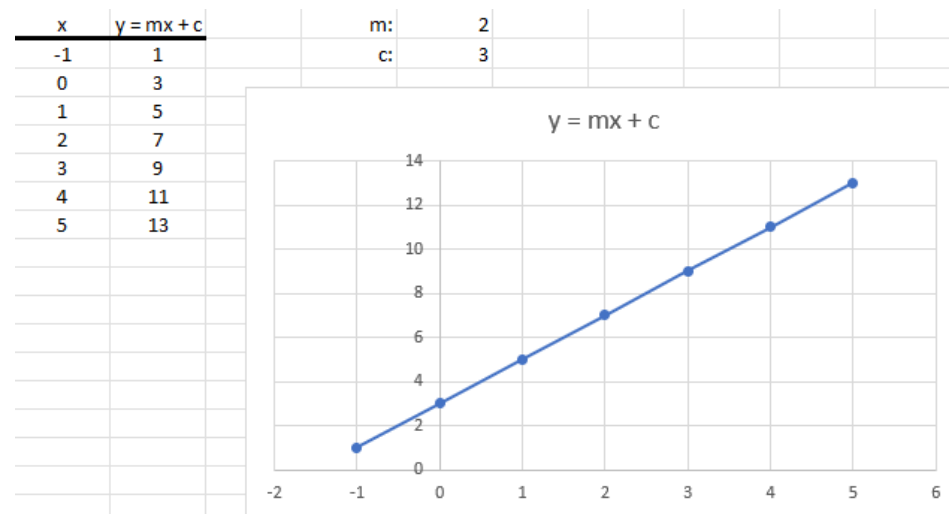


Linear Regression



EQUATION OF A LINE AND TERMINOLOGY

All straight lines have an equation that can be written in the form $y = mx + c$.



m is referred to as the **coefficient of x**. On a plot we see it's effect as the **gradient** of the line. For data analytics, the symbol β_1 might be used instead. We can interpret it as the impact that changing x has on the predicted y value – if x increases by 1 unit, m is the amount that the y prediction changes by.

c is a **constant**. On a plot it controls the height of the graph. When $x=0$ we see that it is where the line **intercepts the y-axis**. For data analytics, the symbol β_0 might be used instead.



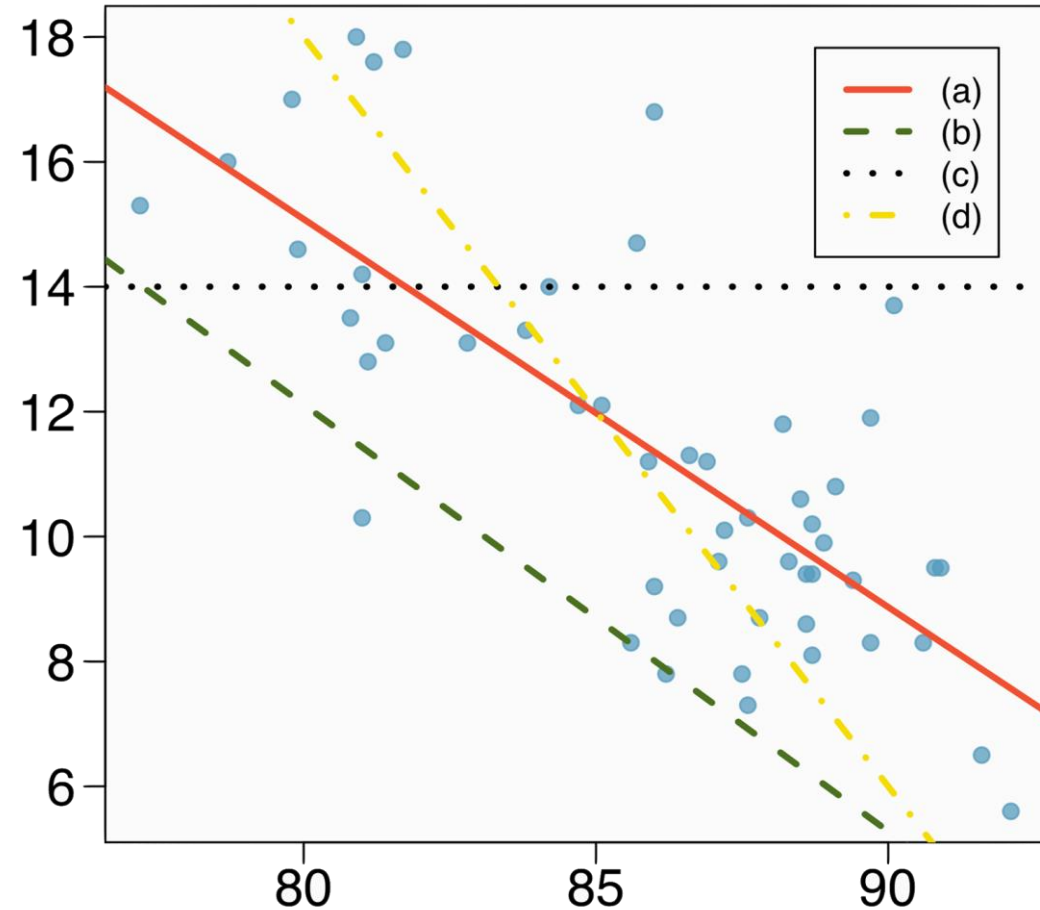
**Residuals
(or Loss)**



EYEBALLING THE LINE OF BEST FIT



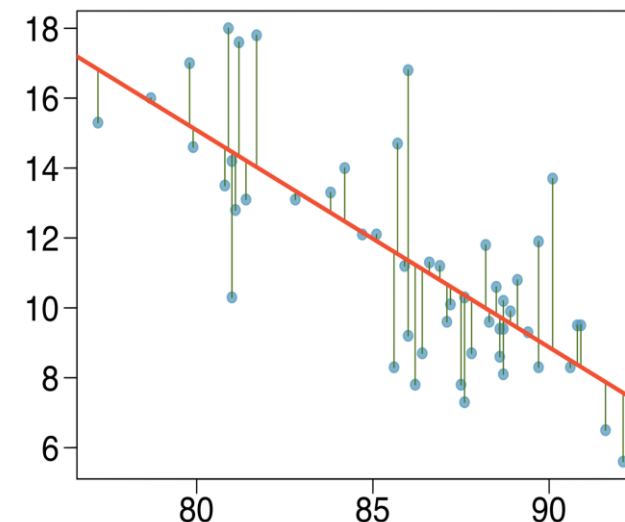
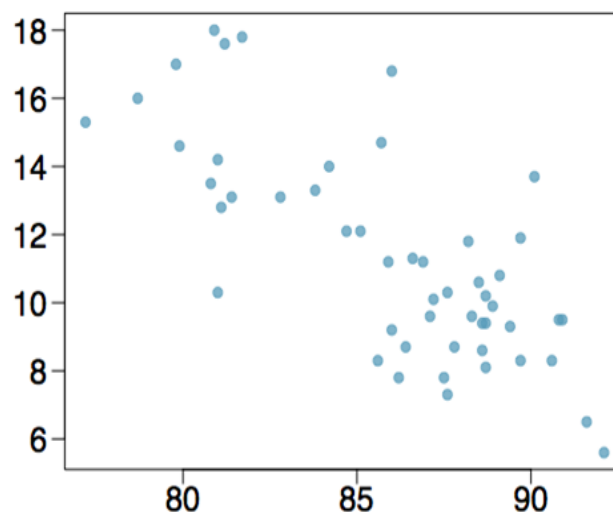
Which of the following appears to be the
line that best fits the data?





RESIDUALS

Real data is rarely located in perfect straight lines!



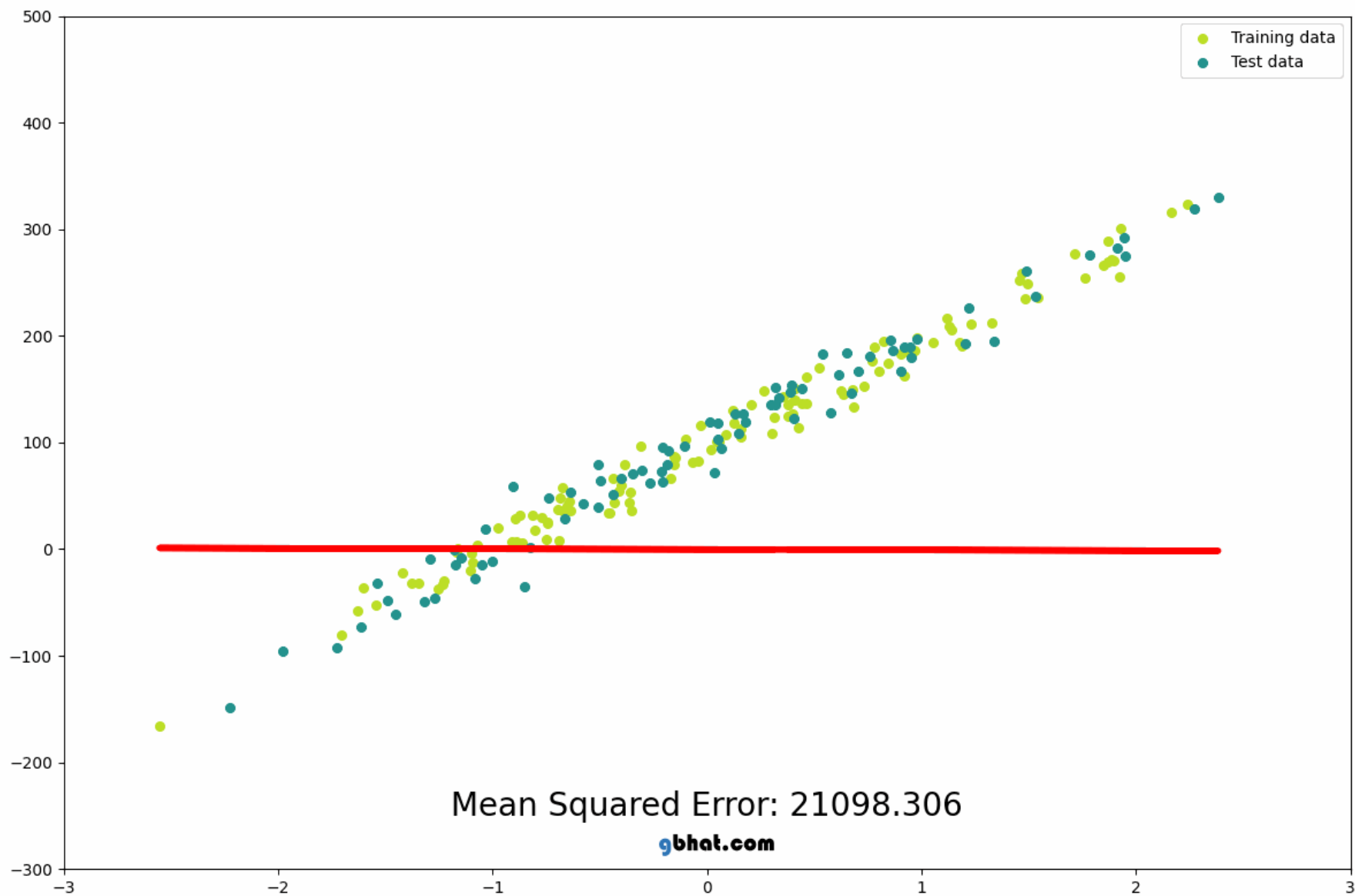
If we fit a line through the data, the **vertical differences** between the data and the line are called **residuals**.

Each one of these residuals shows the difference between the real y-coordinate and what the equation of the line predicts from the x coordinate.

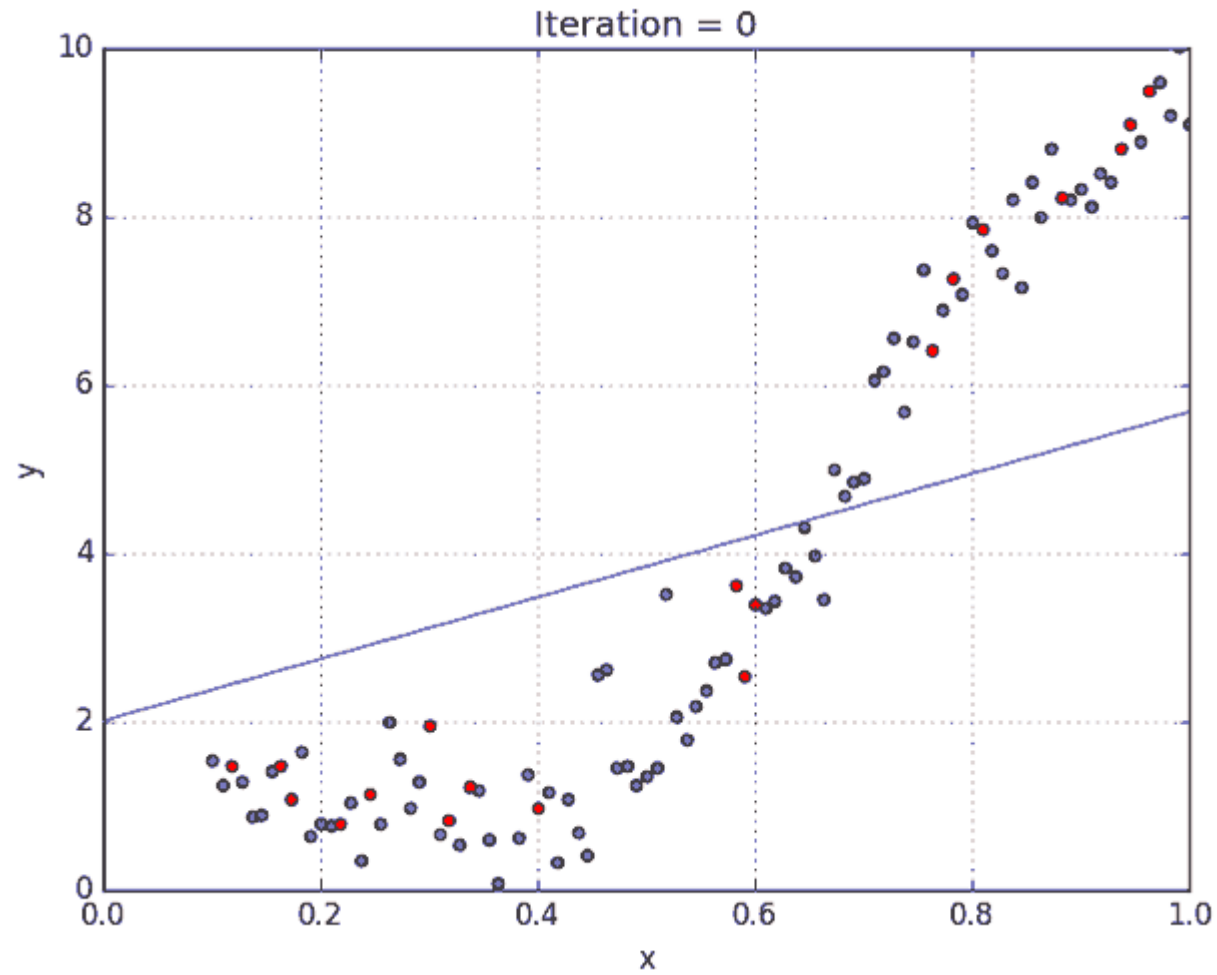
Residuals represent the **error** that a prediction would have. **Linear Regression** is a method that involves finding the best β_1 and β_0 (i.e. the best equation $y = mx + c$) to minimise the error.



QA Regression coefficients



QA Regression coefficients





RESIDUALS

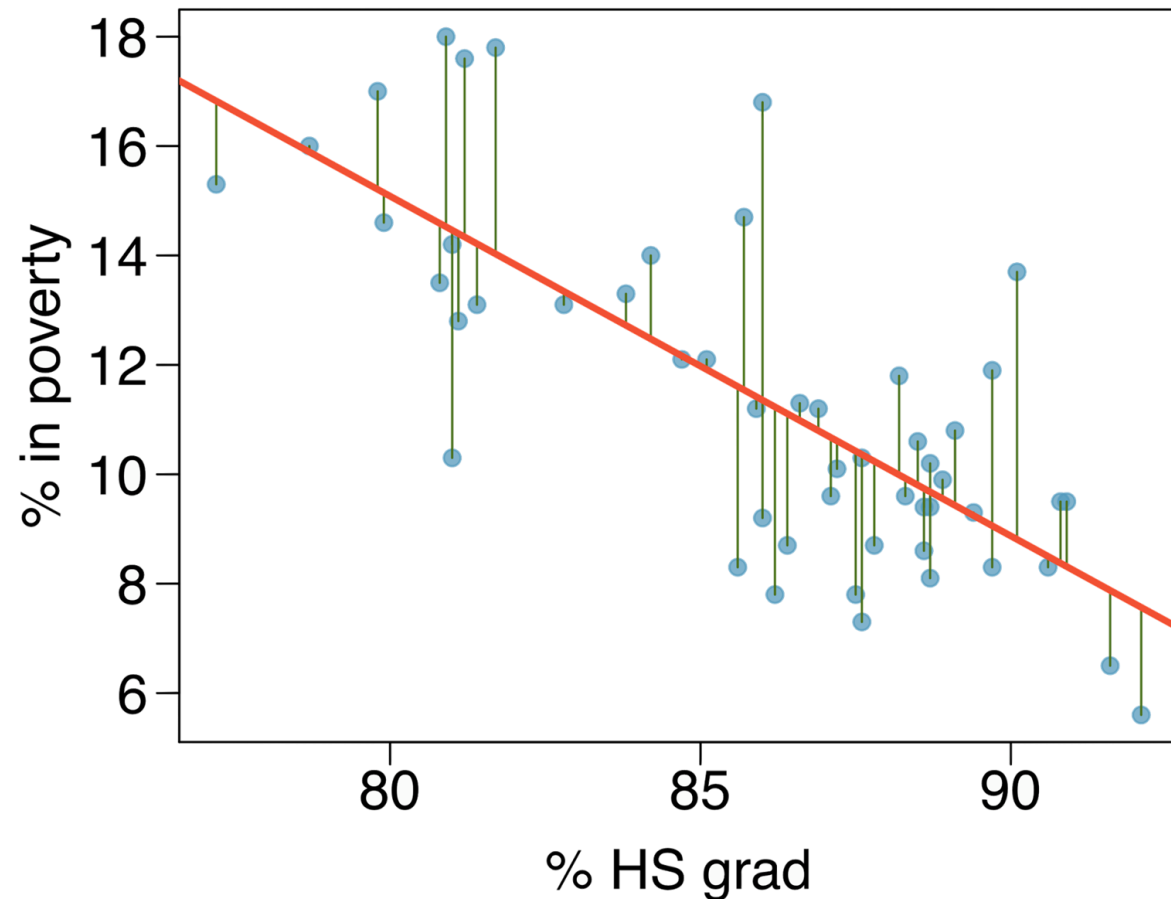


Leftovers from the model fit:

Data = Fit + Residual

Predicted: $\hat{y}_i = mx_i + c$

Observed: $y_i = mx_i + c + e_i$



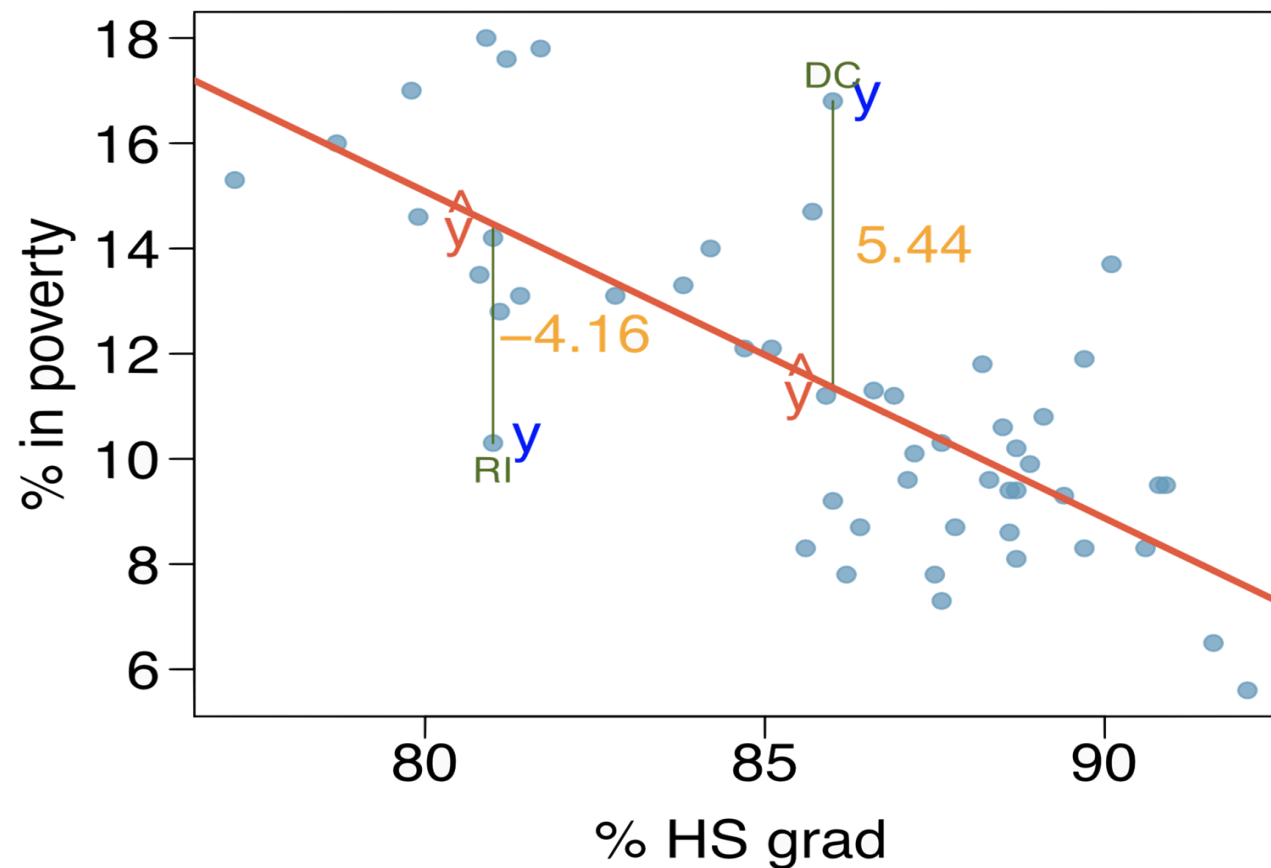


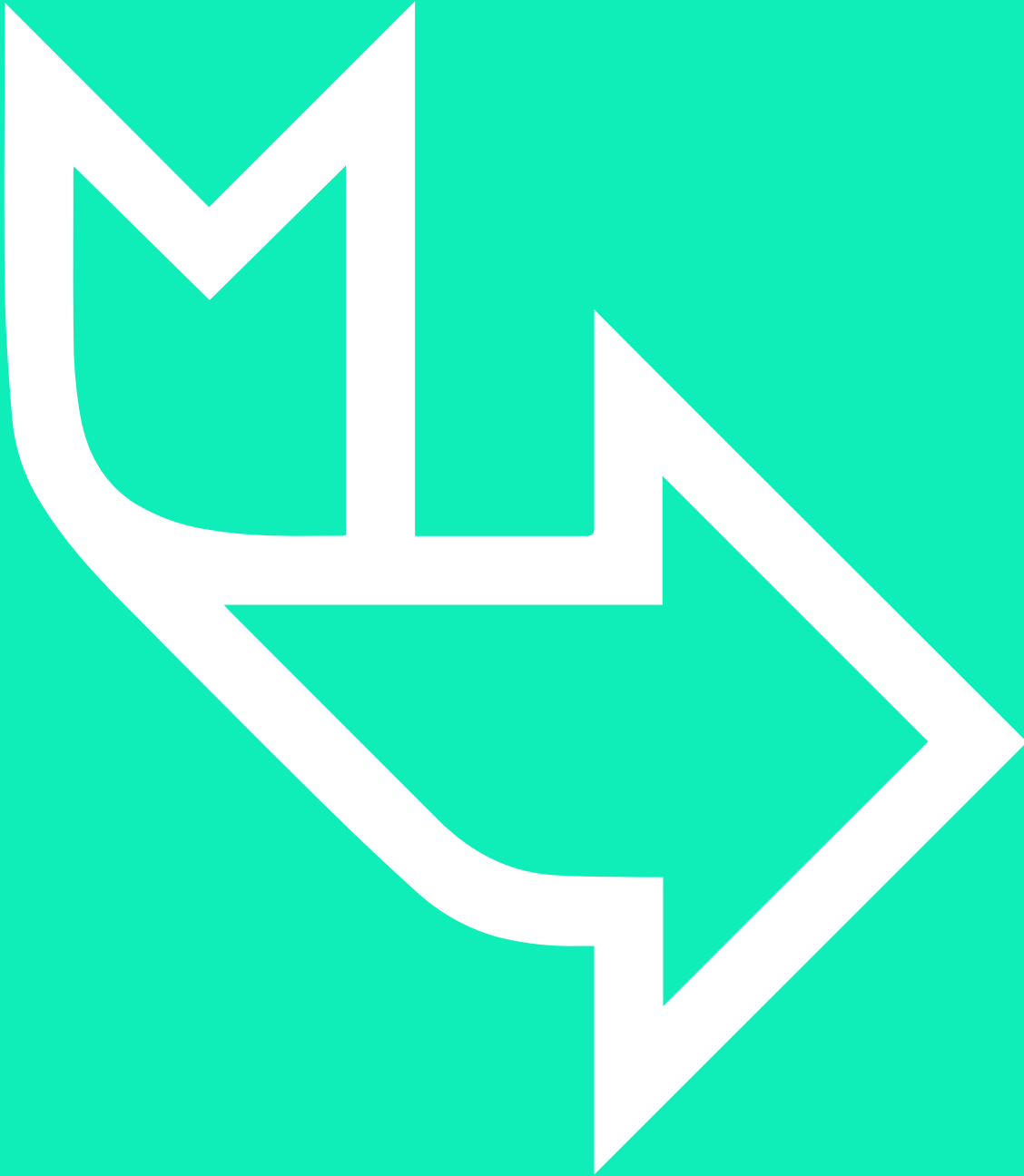
RESIDUALS

Difference between the observed (y_i) and predicted $\hat{y}_i$

$$e_i = y_i - \hat{y}_i$$

% living in poverty in RI is 4.16% less than predicted





Assignment: Regression Modelling

Download and open the
Training_Regression.ipynb
notebook



Fitting a Regression Model



FITTING

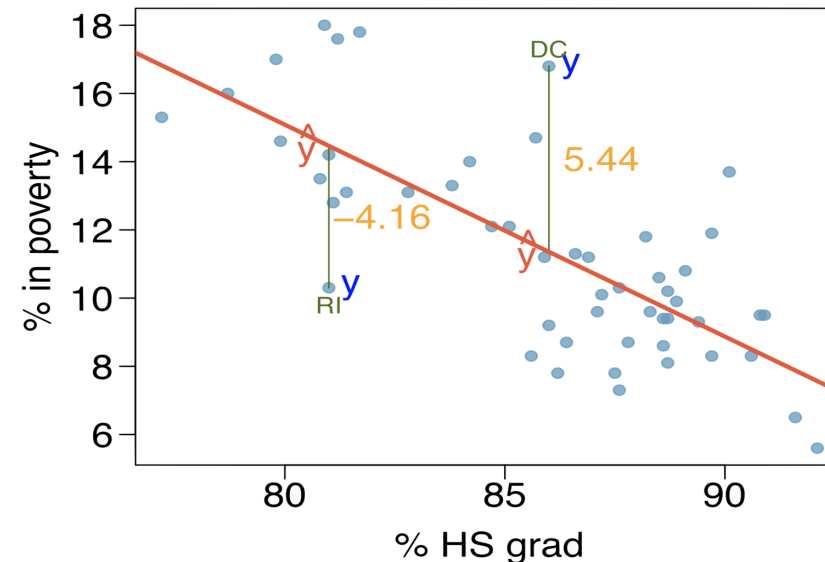
We want a line that has small residuals...

Option 1: Minimise the sum of magnitudes (absolute values) of residuals

$$|e_1| + |e_2| + \dots + |e_n|$$

Option 2: Minimise the sum of squared residuals – ‘least squares’

$$e_1^2 + e_2^2 + \dots + e_n^2$$





FITTING

We want a line that has small residuals...

Option 1: Minimise the sum of magnitudes (absolute values) of residuals

$$|e_1| + |e_2| + \dots + |e_n|$$

Option 2: Minimise the sum of squared residuals – ‘least squares’

$$e_1^2 + e_2^2 + \dots + e_n^2$$

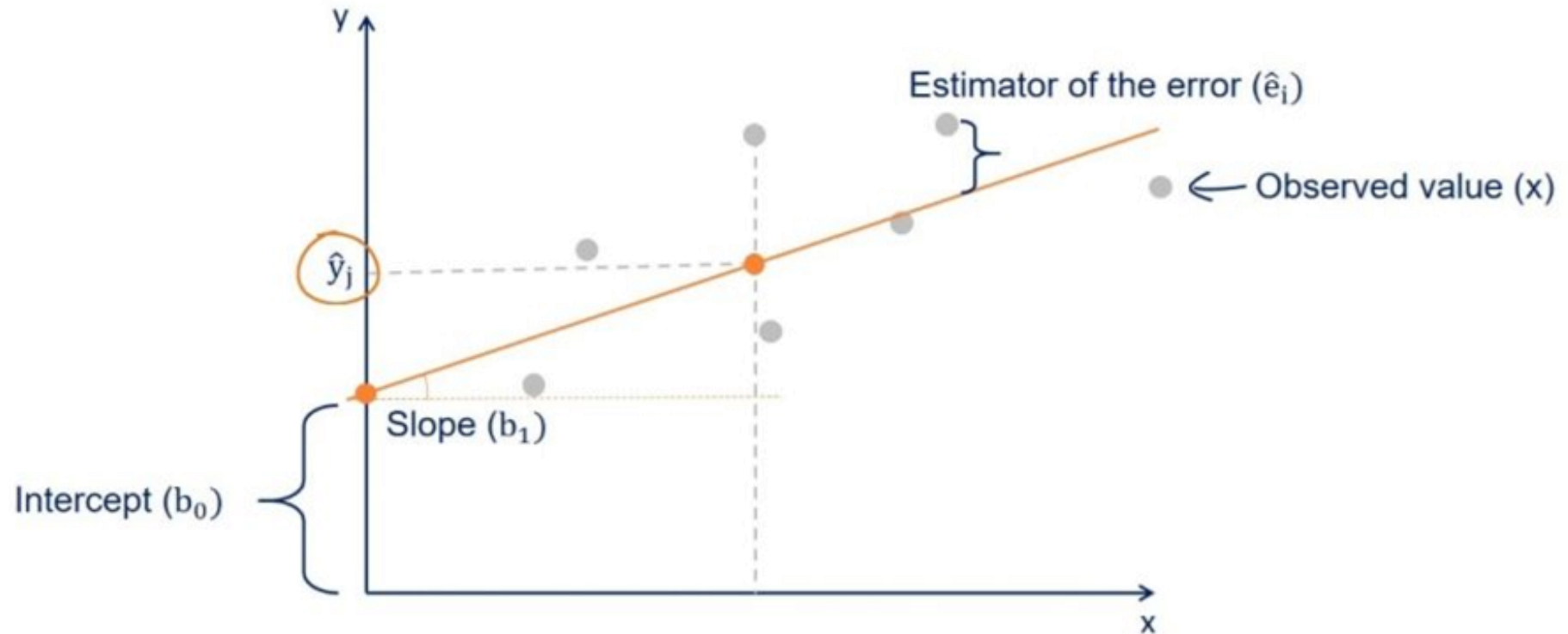
Why least squares?

1. Most commonly used.
2. Easier to compute by hand and using software.
3. In many applications, a residual twice as large as another is usually more than twice as bad.



QA Regression coefficients

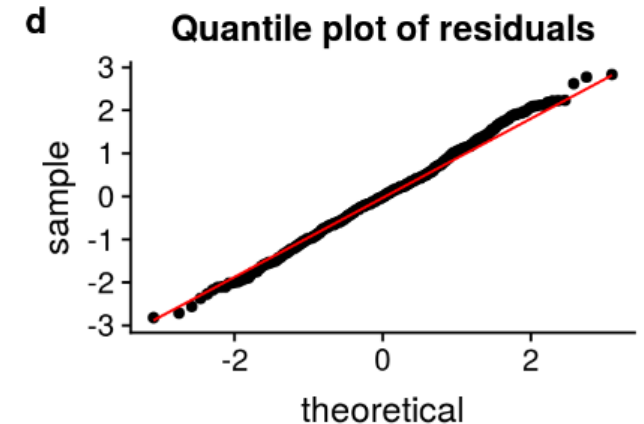
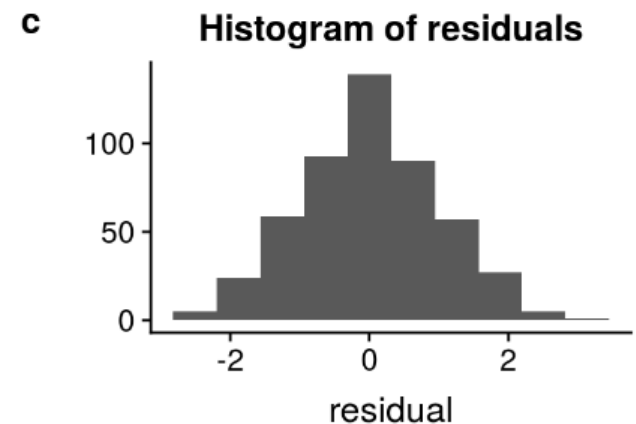
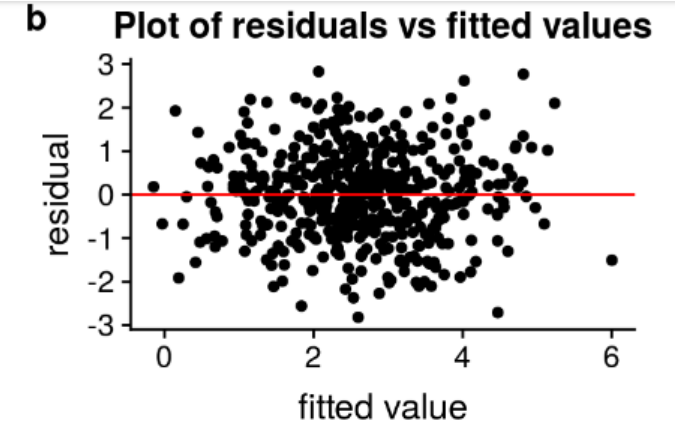
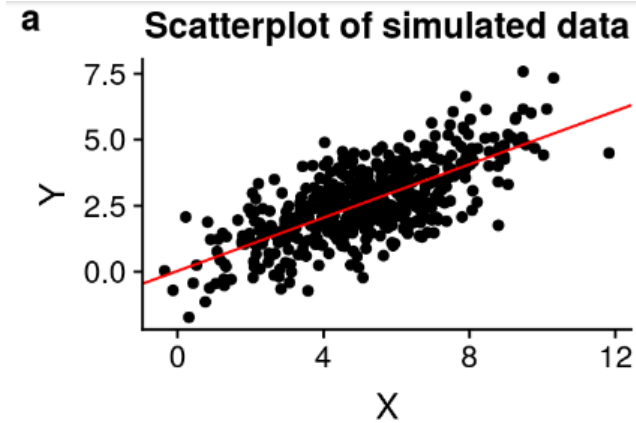
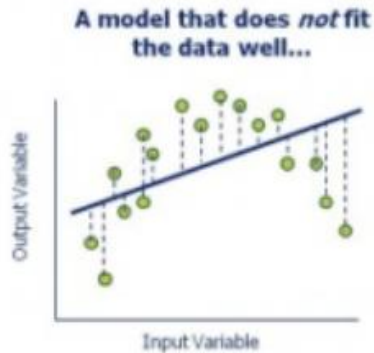
$$Y_i = \beta_0 + \beta_1 X_i + \hat{\epsilon}_i$$





Least squares Regression Line: Assumptions

1. **Linearity**
2. **Nearly normal residuals**
3. **Constant variability**



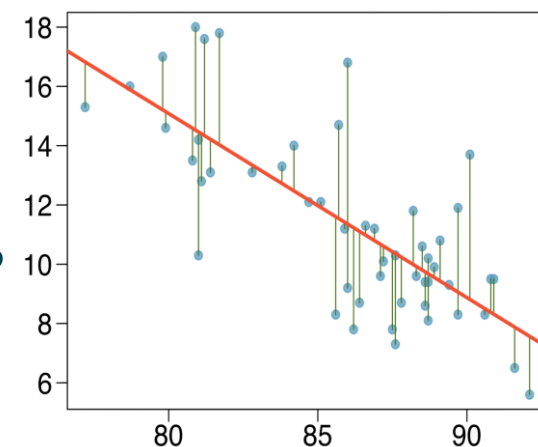


RESIDUALS LEARNING CHECK

1. What are Residuals
2. How are they used to fit our model to the data?
3. Why do we square the errors?
4. What do each of these elements represent?

$$Y_i = \beta_0 + \beta_1 X_i + \hat{\epsilon}_i$$

5. What are the conditions that must be fulfilled when fitting a line using Least Squares method?





Evaluating Regression



REGRESSION METRICS: ERRORS

Root Mean Squared Error (RMSE)

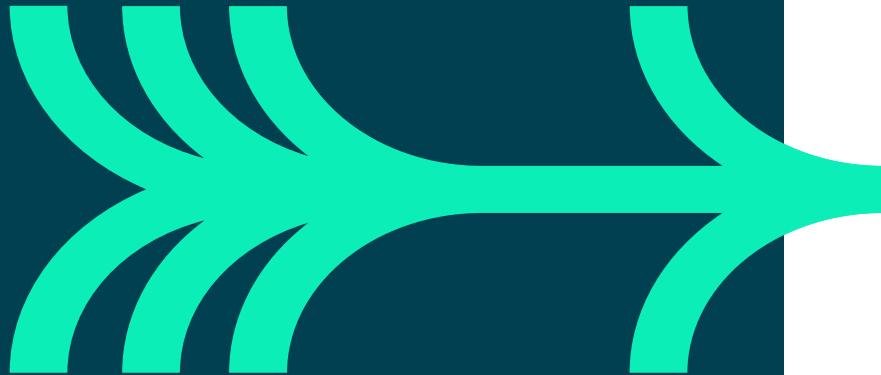
$$RMSE = \sqrt{\frac{1}{N} \sum_{i=0}^N (y_i - \hat{y}_i)^2}$$

Mean Absolute Error (MAE)

$$MAE = \frac{1}{N} \sum_{i=0}^N |y_i - \hat{y}_i|$$

There are others!

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

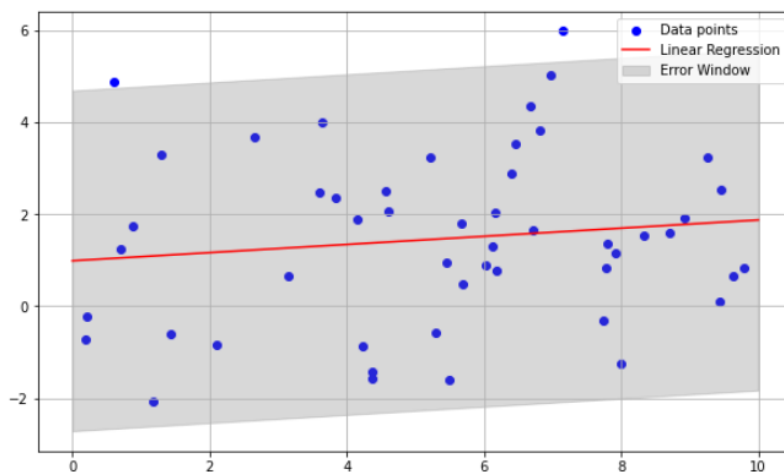




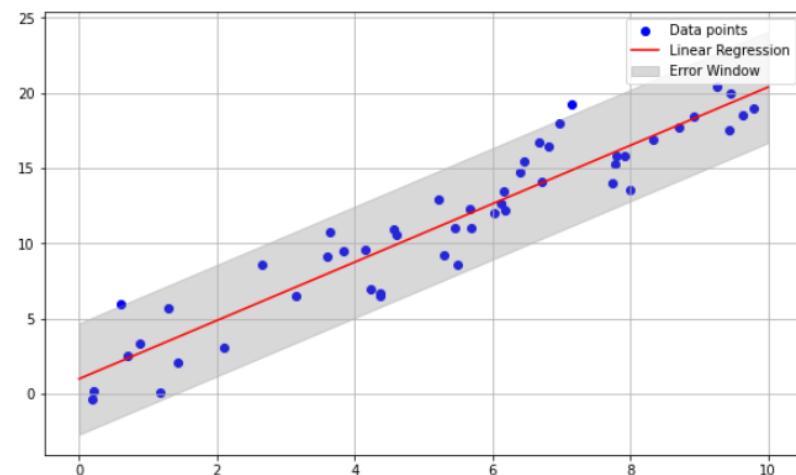
REGRESSION METRICS: USEFULNESS

R^2 is the **coefficient of determination**. It can be used as a measure of **usefulness** of the model. It's a measure of the variance of the dependent variable, y . It's on a scale of 0 (bad) to 1 (good).

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

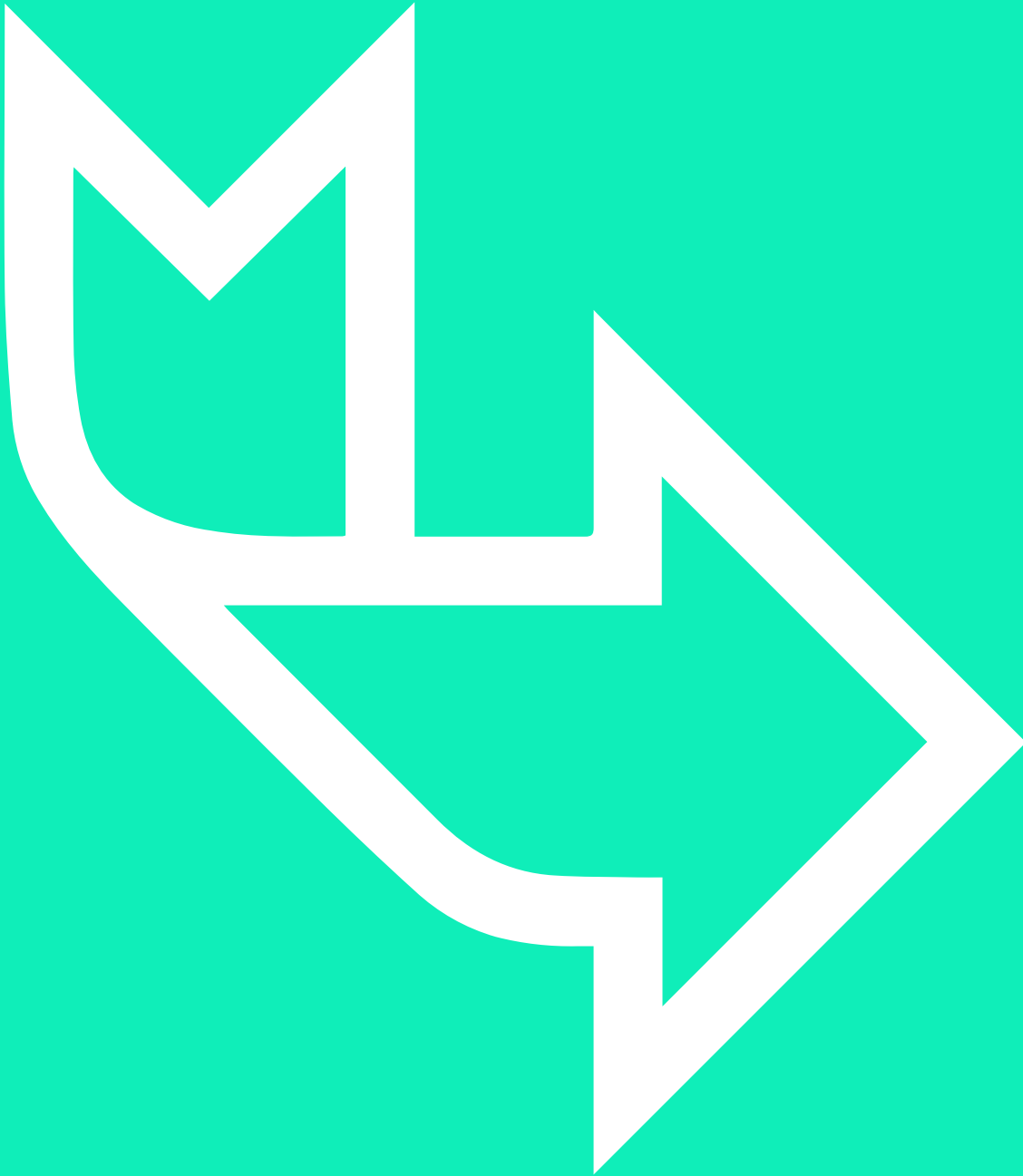


R Squared = 0.02



R Squared = 0.89

Consider the predicted y values in these cases – if you predict from different x inputs, does it make much difference to what is predicted?



Assignment: Regression Modelling

Continue working on the
Training_Regression.ipynb
notebook



Supervised Machine Learning: Classification



Logistic Regression

QA Calculating Risk for Classification

Classification models that guide decision making and automated processes. What trait do these classifications all share?

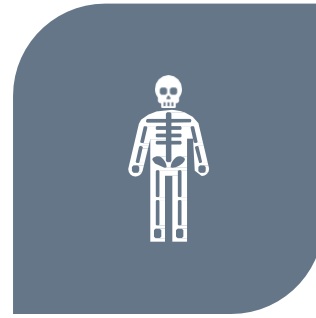


Spam Detection:

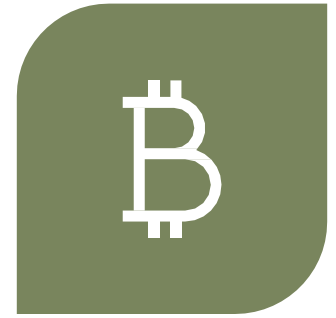
Is this email Spam?



Credit Card Fraud:



Healthcare:



Banking:



WHAT IS LOGISTIC REGRESSION?

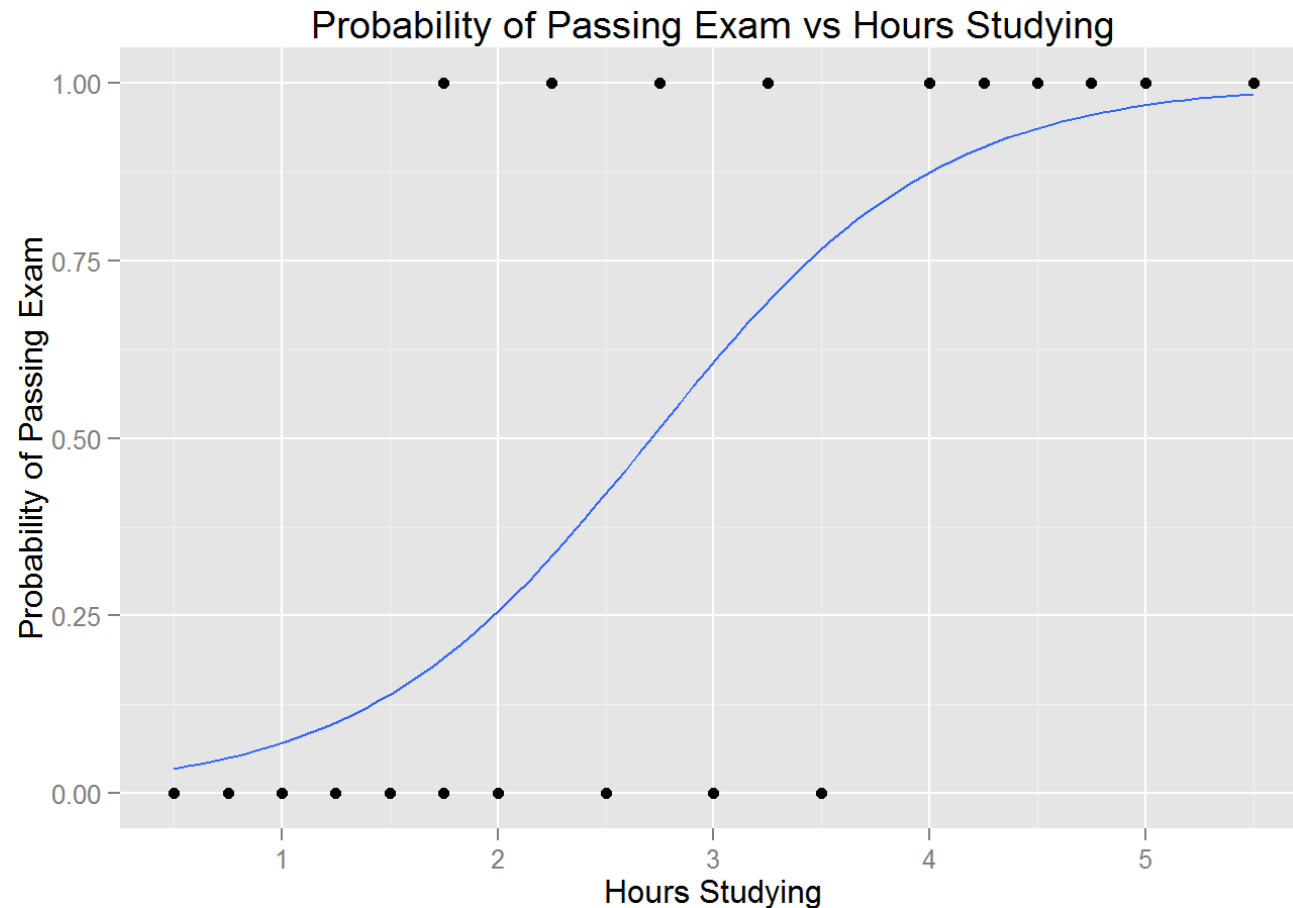


- Logistic regression is an extremely popular **machine learning** technique.
- Despite the name, logistic regression is a **classification** algorithm.
- The output of a logistic regression model is the **probability** of an occurrence belonging to a **particular class**, e.g., classifying emails as being spam or not.
- In our case study, we will be using it as a **binary** classifier, although it can also be applied to **multinomial** and **ordinal** classes

QA Binary Classification - Logistic Model

Hours Studied	Exam pass/fail
0.25	Fail
0.7	Fail
1	Fail
1.25	Fail
1.5	Fail
1.75	Pass
2	Fail
2.2	Pass
2.5	Fail
3	Fail
3	Pass
3.5	Fail
4	Pass
4.2	Pass
4.5	Pass
5	Pass

Given the dataset, notice how a **logistic curve** can be used to predict probability of a binary outcome.



QA What is logistic regression

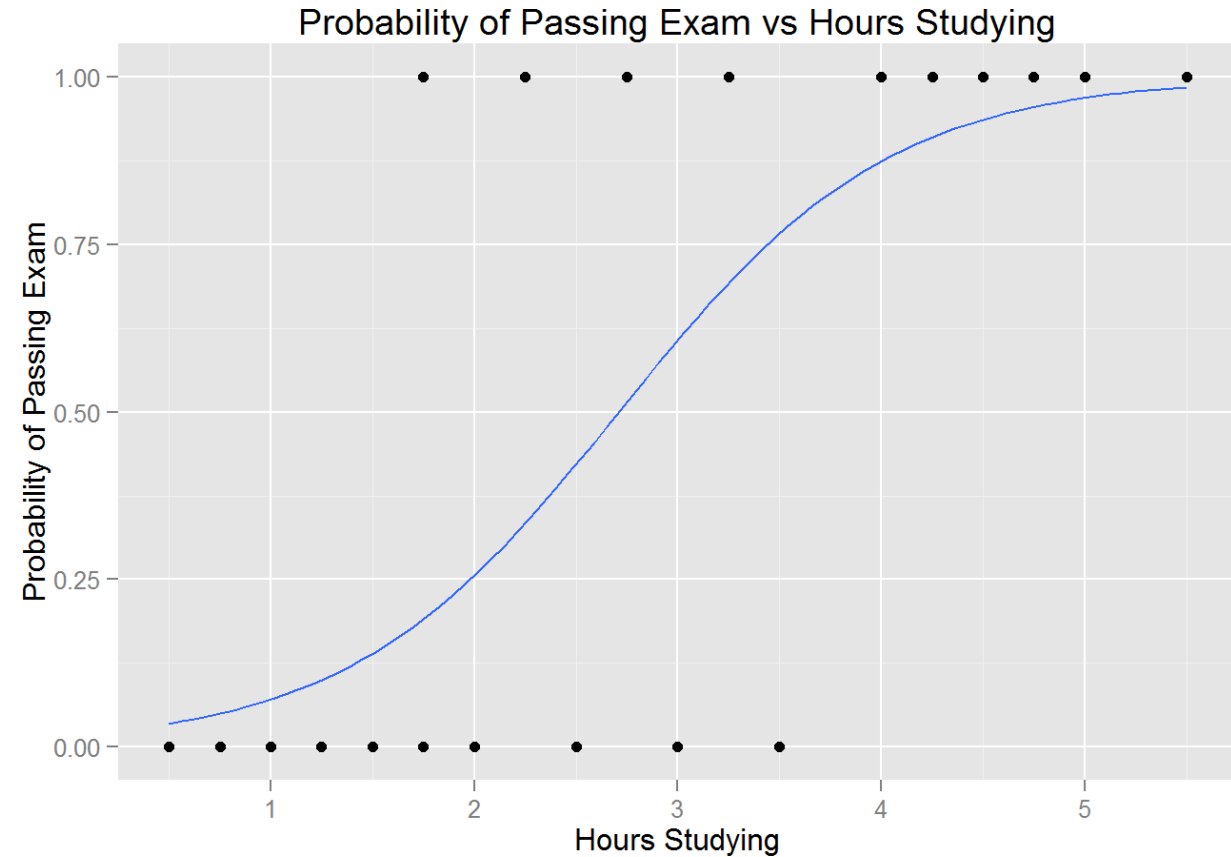
Logistic regression is used to estimate the likelihood of an outcome based on the input variables.

For our binary outcome we set a threshold at 0.5, this can be changed during model tuning.

Can be used as a classifier, with the most likely output chosen.

- I. **Input** variables can be continuous or discrete.
- III. **Output** : A model with Coefficients of each variable

A linear expression which predicts the log-odds ratio between 0 and 1 (unlike a linear mode which must have a continuous outcome variable)



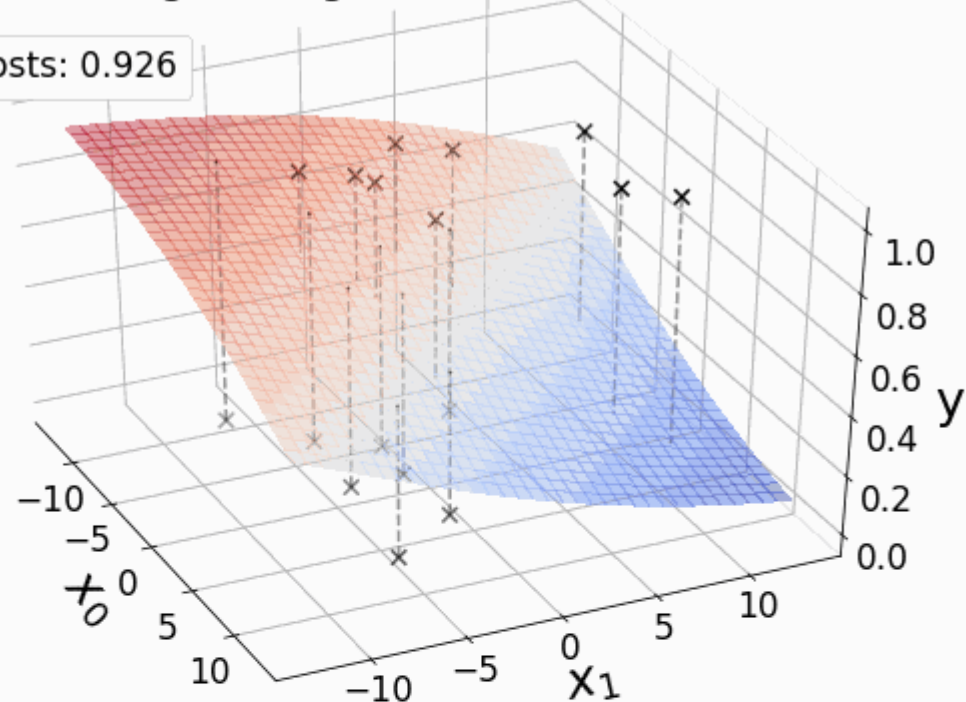


Training a Logistic model

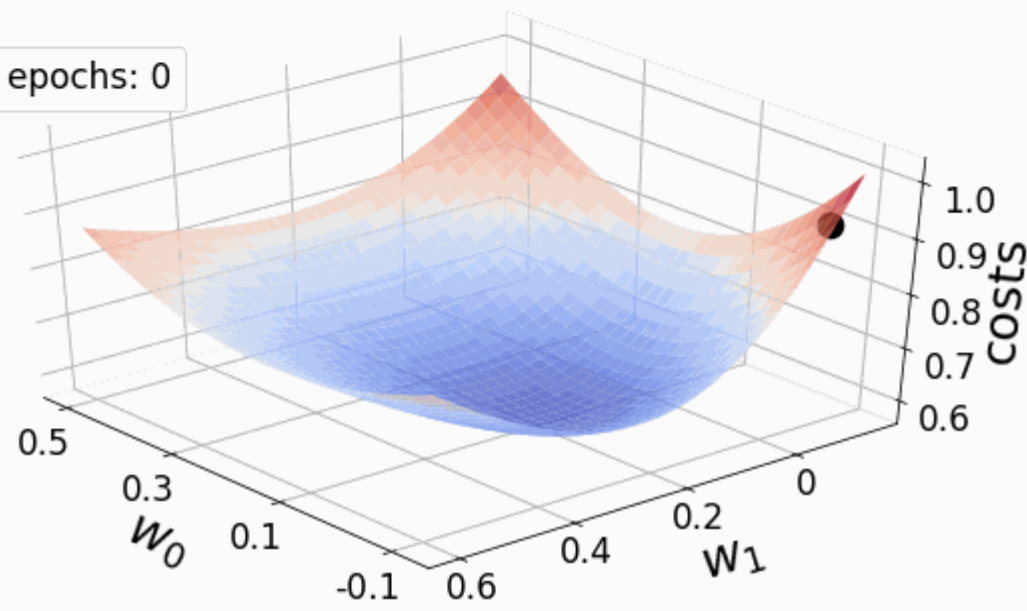
- Starts as a linear model first

Logistic regression curve (3D)

----- costs: 0.926



----- epochs: 0





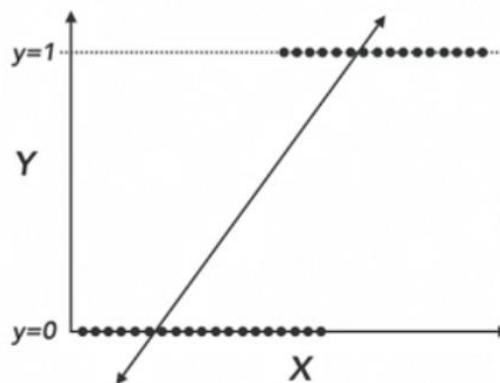
THE MODEL

Here is a simple example with one **support variable** (hours studying) and a **binary target** (pass or fail an exam).

As opposed to **linear regression**, which builds a **straight line**, **logistic regression** builds a **sigmoid function** that gives us the **probability** of a **data point** belonging to a particular class.

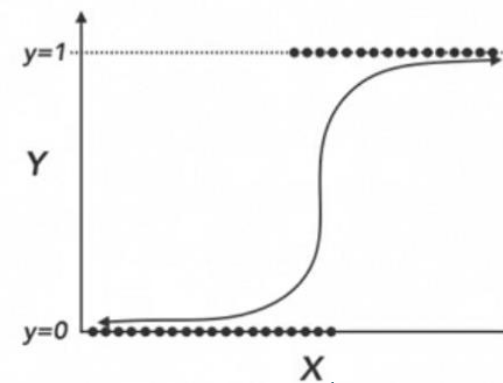
$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Linear Regression



$$Y = \beta_0 + \beta_1 x$$

Logistic Regression



$$Y = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Source



THE MATHS

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

As the output of the logistic regression is a probability, it will always be between 0 and 1.

$$\text{As } (\beta_0 + \beta_1 x) \rightarrow \infty \quad 1 + e^{-(\beta_0 + \beta_1 x)} \rightarrow 1 \quad p(x) \rightarrow 1$$

$$\text{As } (\beta_0 + \beta_1 x) \rightarrow -\infty \quad 1 + e^{-(\beta_0 + \beta_1 x)} \rightarrow 0 \quad p(x) \rightarrow 0$$

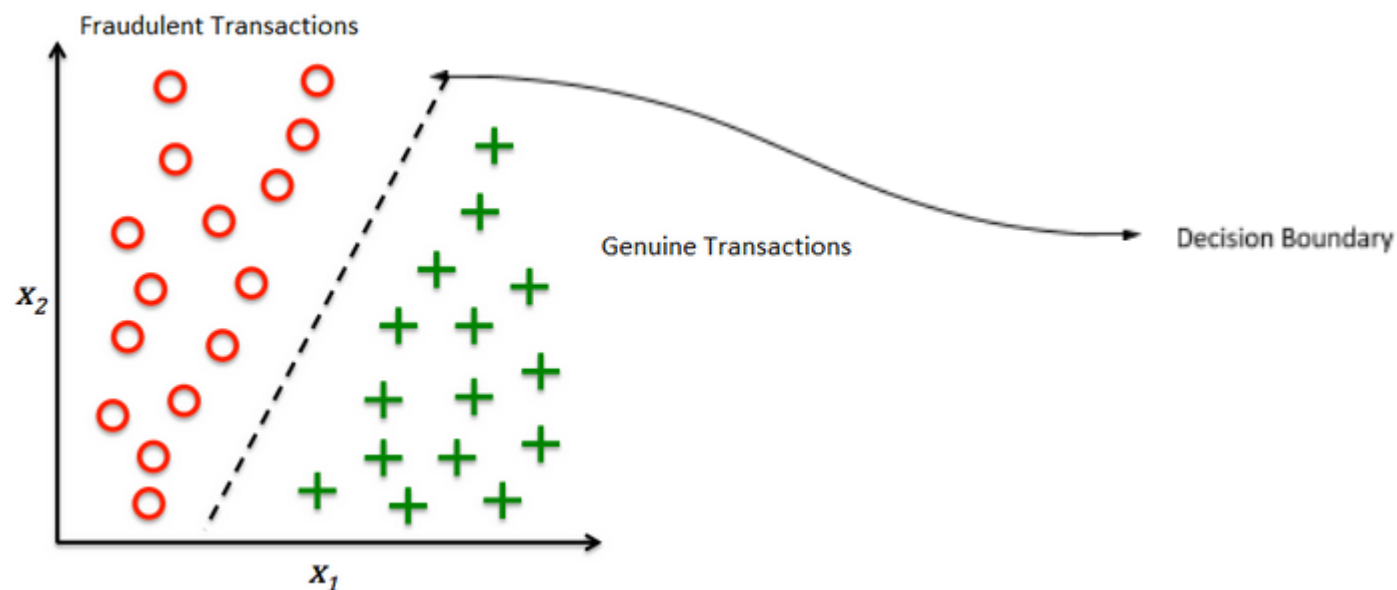
$\beta_0 + \beta_1$ are discovered during the machine learning process and are the result of finding the parameters of the **hyperplane** that best separates the data.

Logistic regression is regarded as a **linear model**, as the parameters for the model are a linear combination.



ASSUMPTIONS

One of the key assumptions for standard logistic regression is that the **binary groups** are **linearly separable**, or **mostly linearly separable**, i.e., not perfectly separable, as shown below:



separable and balanced data, meaning we have an **equal distribution of classes**.

Source

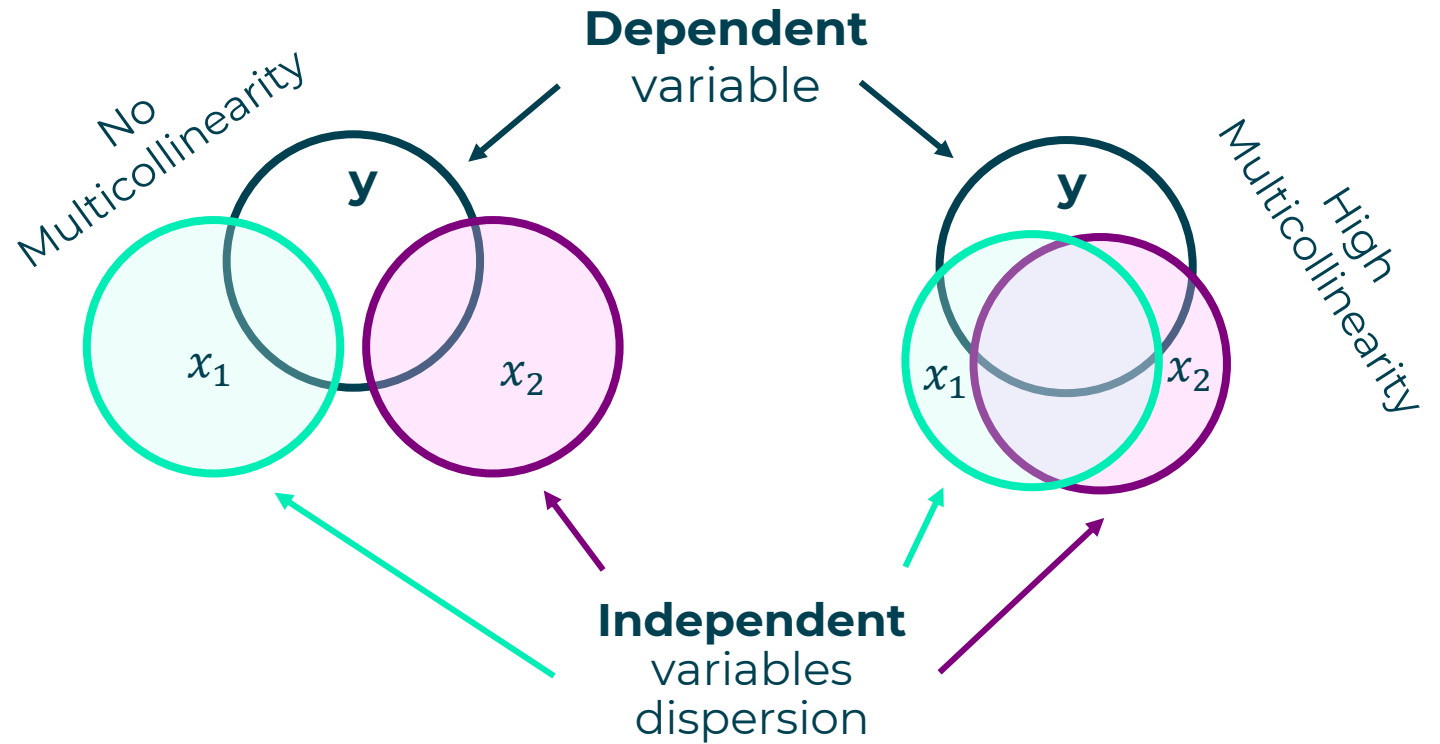


ASSUMPTIONS:

ALL SAMPLES
ARE
INDEPENDENT
OF EACH OTHER

Multi-collinearity:

The solutions we have for this in logistic regression are the same as the ones we use for standard regression cases. **Investigate with support variables**



$$\hat{y} = b_1 \cdot x_1 + b_2 \cdot x_2 + \dots + b_k \cdot x_k + a$$

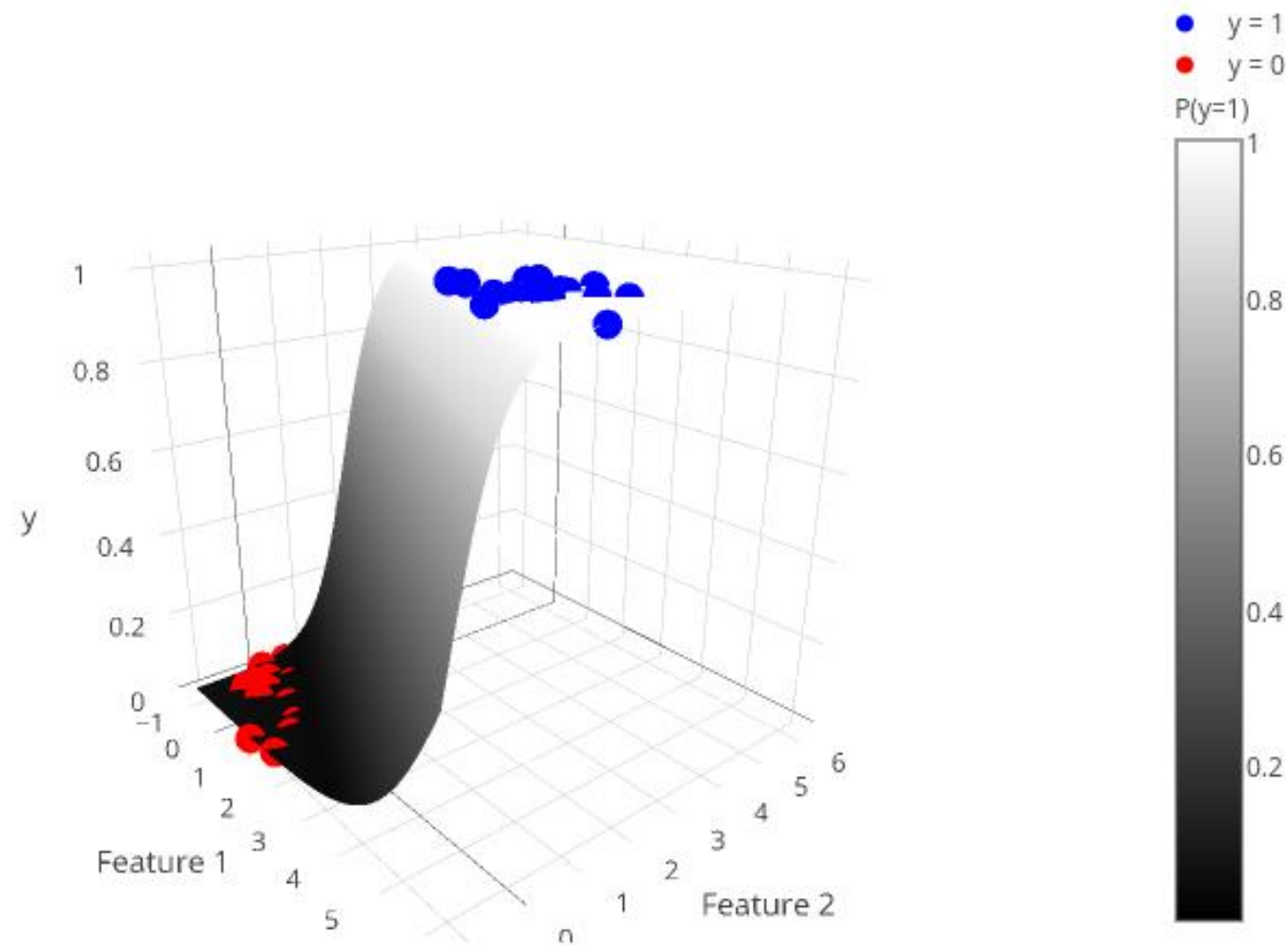
$$\hat{x}_1 = b_2 \cdot x_2 + \dots + b_k \cdot x_k + a$$

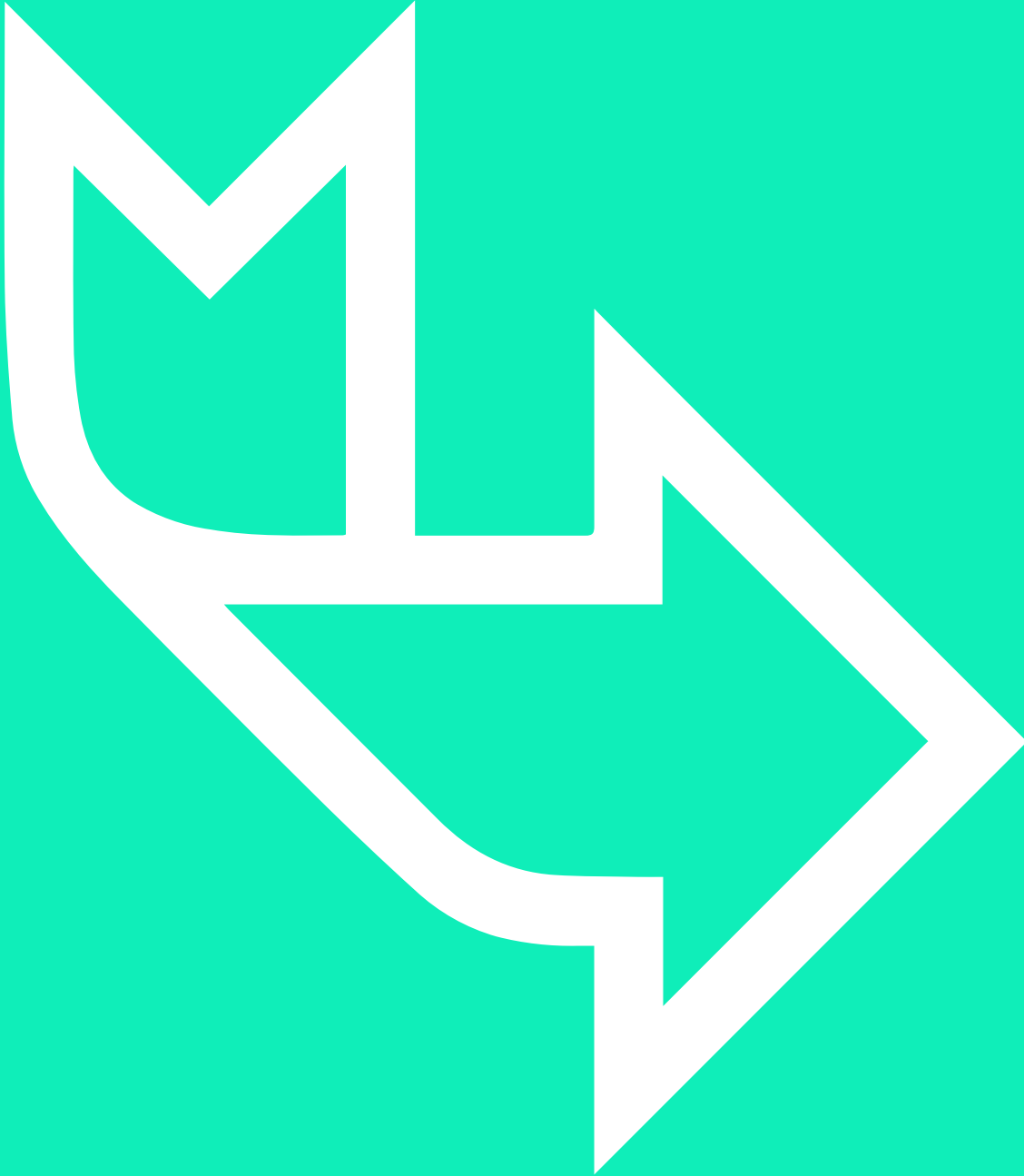
$$\hat{x}_k = b_1 \cdot x_1 + b_2 \cdot x_2 + \dots + a$$

$$\hat{x}_2 = b_1 \cdot x_1 + \dots + b_k \cdot x_k + a$$



LOGISTIC REGRESSION WITH TWO SUPPORT VARIABLES





Assignment: Classification Modelling

Work through the Tasks in the notebook, until you have completed building your first model.



Decision Trees



DECISION TREE



Decision trees can be used for both classification and regression.

A decision tree is a flowchart-like structure in which each internal node represents a 'test' on an attribute (e.g., whether a coin flip comes up heads or tails).

Each branch represents the outcome of the test, and each leaf node represents a class label or an interval, if using for regression.

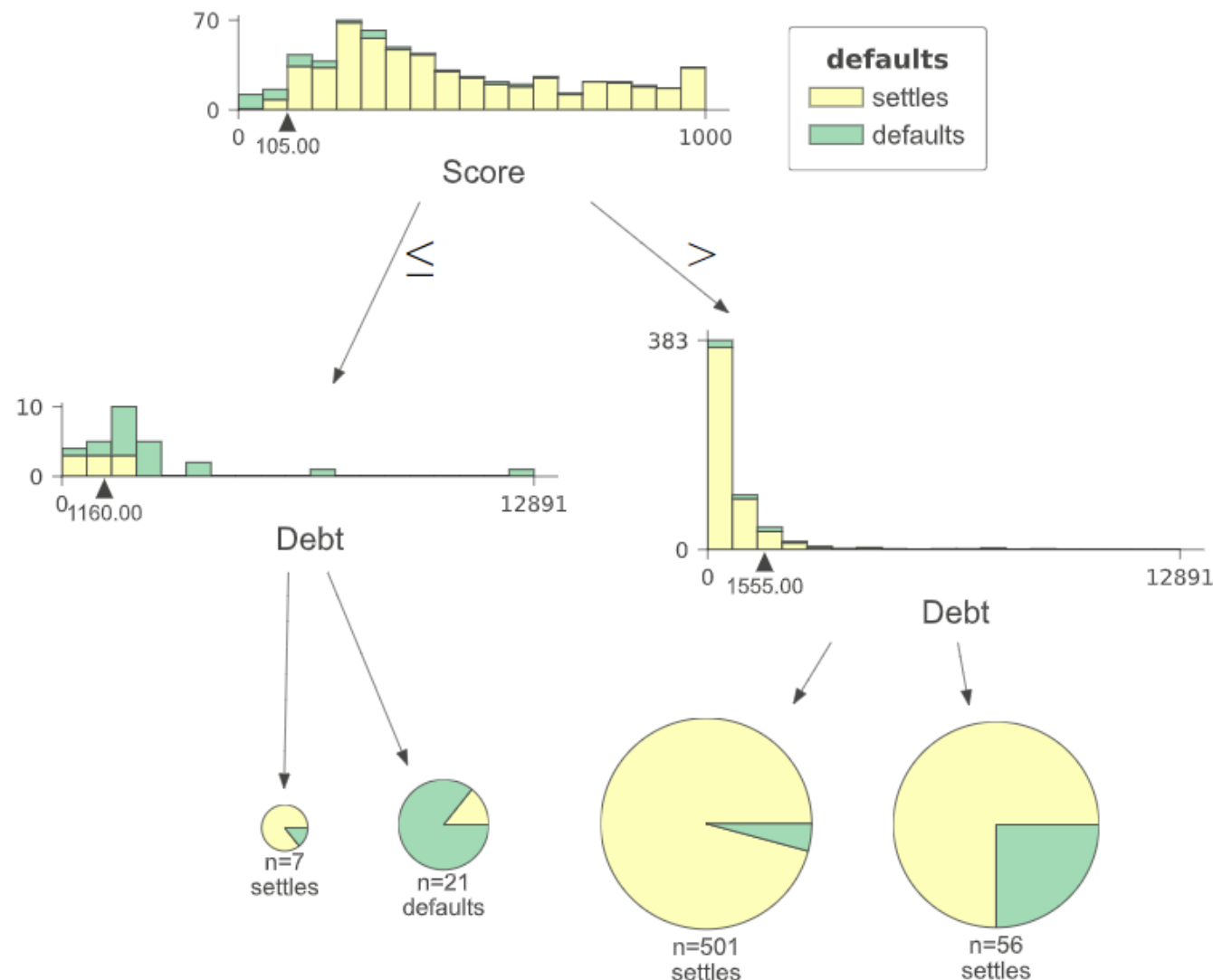


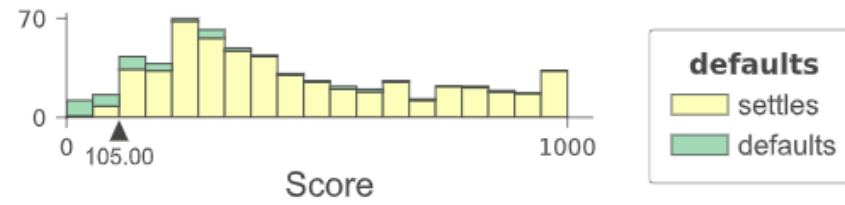
Decision tree for classification

An illustration of a decision tree on the loans data.

This tree has a max depth of 2.

Max depth is the amount of 'questions' we can ask of the data.





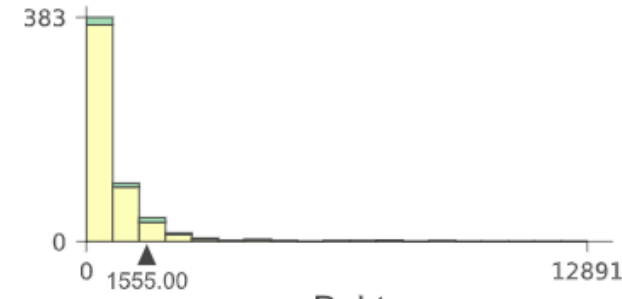
Score

$\leq$

$>$



Debt



Debt



Score

n=2
settles

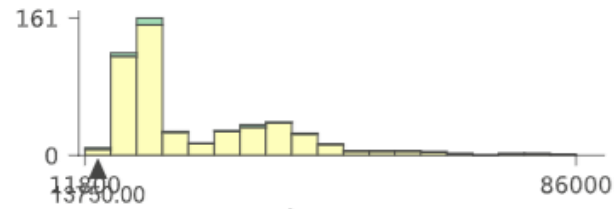
n=5
settles



Balance

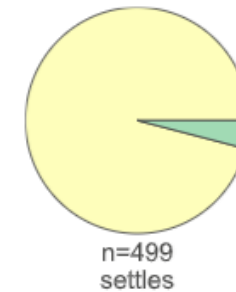
n=15
defaults

n=6
settles

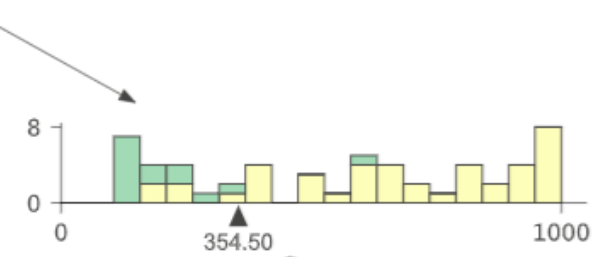


Income

n=2
settles

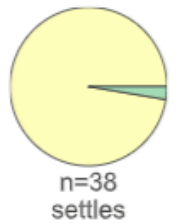


n=499
settles



Score

n=18
defaults



n=38
settles



THE MATHS



We will look at how the Gini Impurity is calculated for decision trees, but there are other metrics that can be used, for example:

Classification:

- Gini Impurity <- Default in Scikit Learn
- Entropy or Information Gain
- Log Loss

Regression:

- Squared Error <- Default in Scikit Learn
- Friedman Mean Squared Error
- Absolute Error

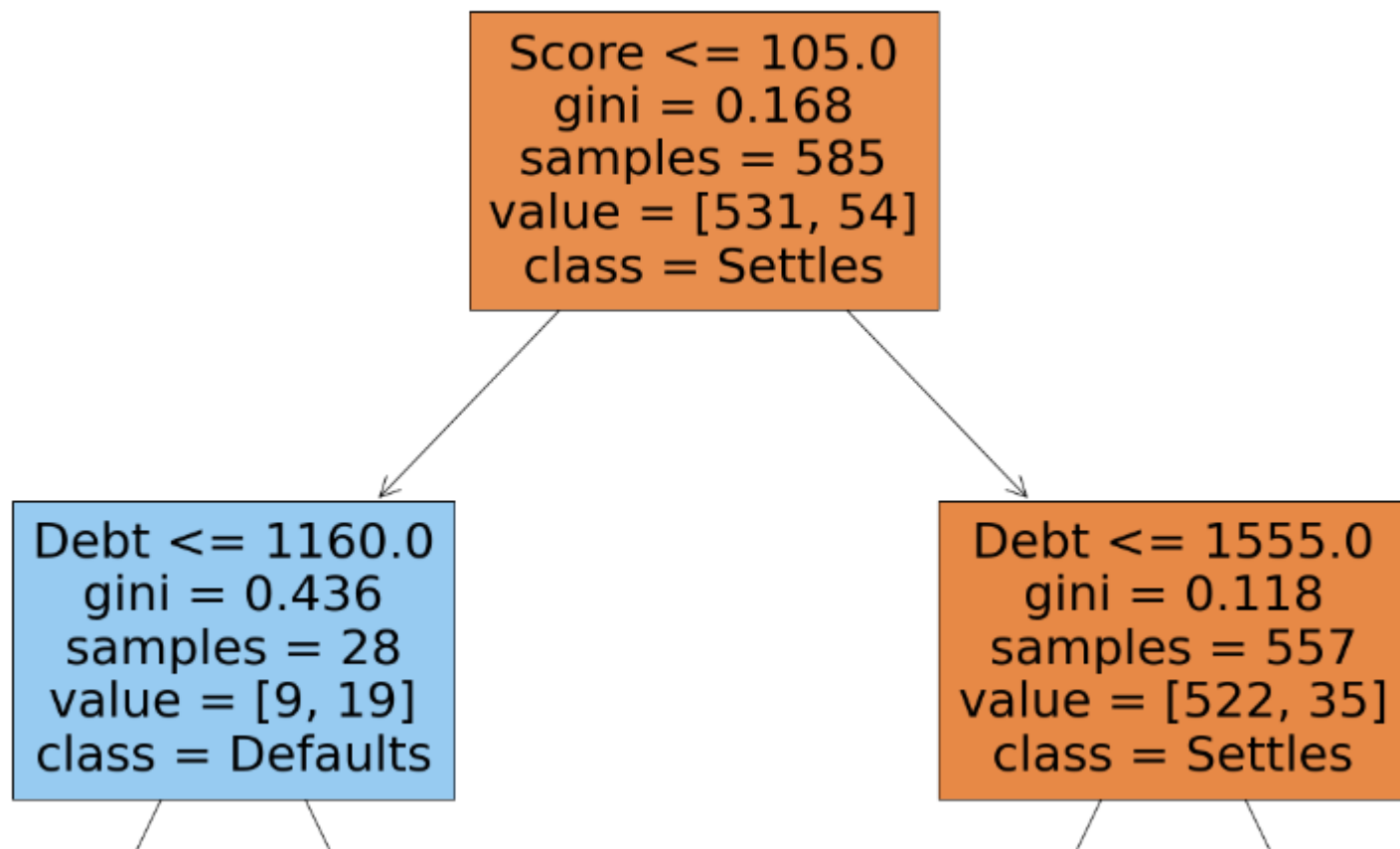
Further reading:
[Scikit-Learn](#)



Application in loans data

Gini impurity calculation

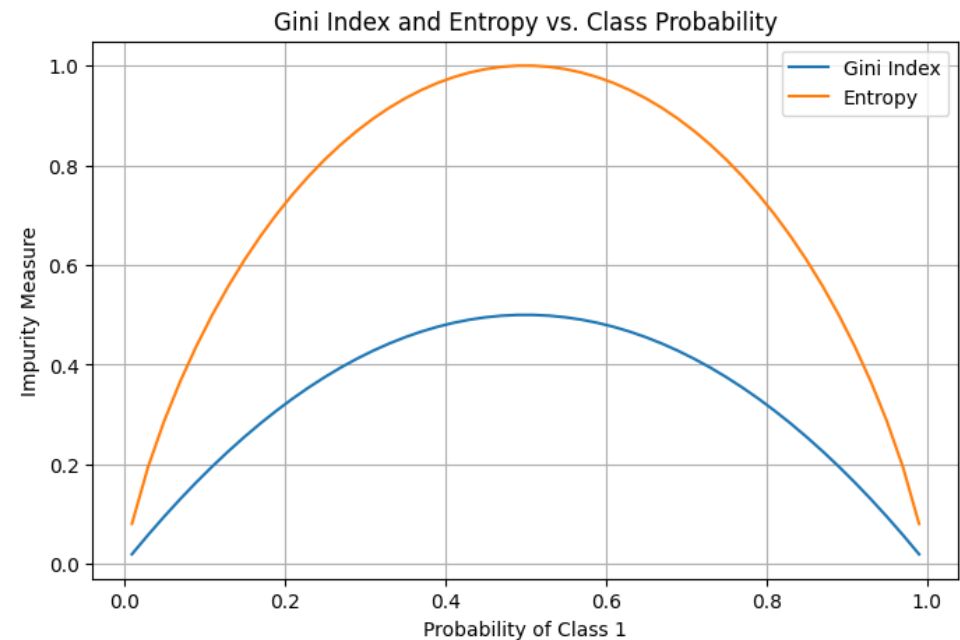
$$G = \sum_{k=1}^K p_k(1 - p_k) = 1 - \sum_{k=1}^K p_k^2$$



The Gini for the first node is calculated:

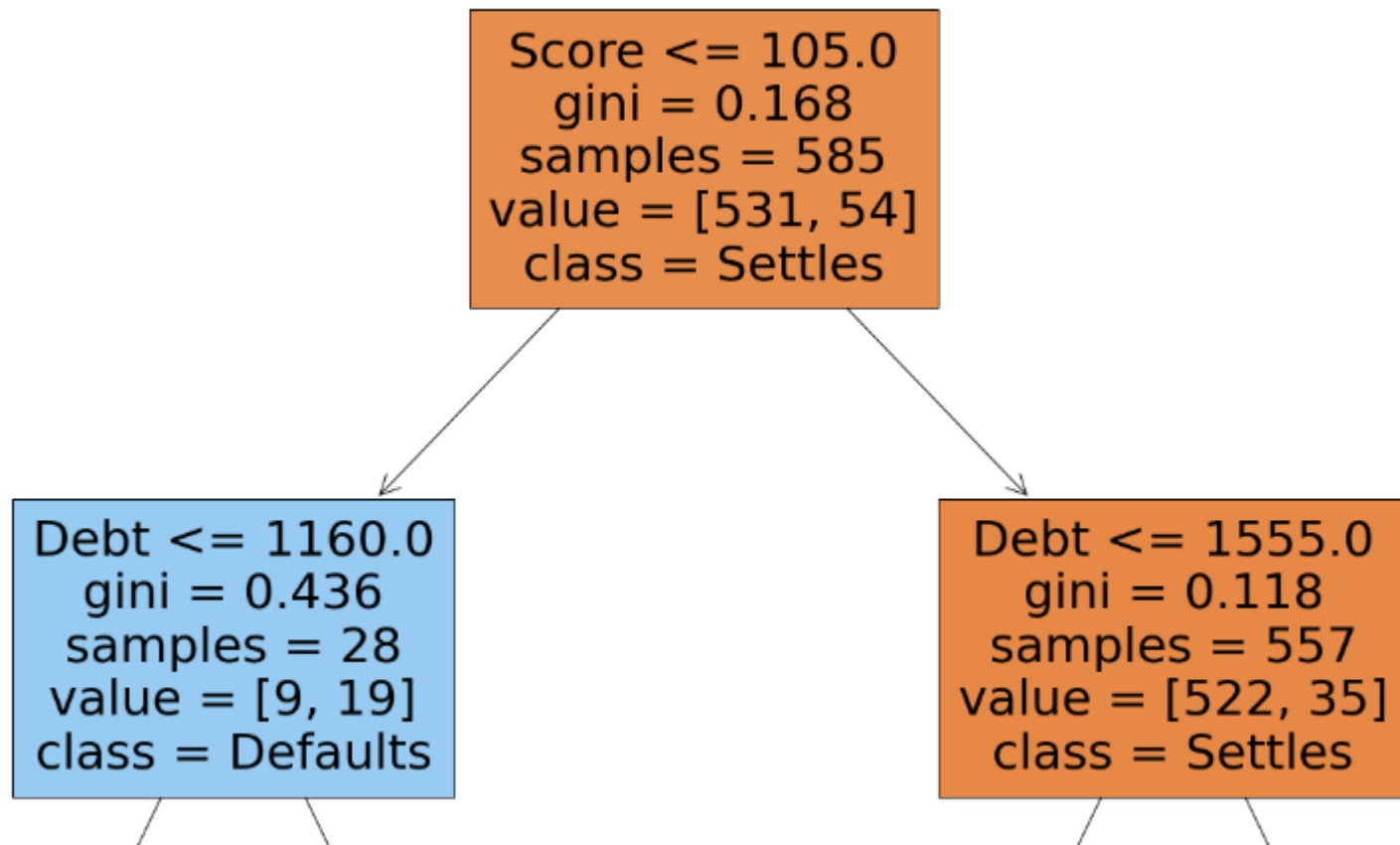
$$\text{Gini} = 2 * (54/585) * (531/585) = 0.168.$$

This is our base line.



Gini impurity calculation

$$G = \sum_{k=1}^K p_k(1 - p_k) = 1 - \sum_{k=1}^K p_k^2$$



The Gini for the first node is calculated:

$$\text{Gini} = 2 * (54/585) * (531/585) = 0.168.$$

This is our base line.

QA Application in loans data

Gini impurity calculation

$$G = \sum_{k=1}^K p_k(1 - p_k) = 1 - \sum_{k=1}^K p_k^2$$

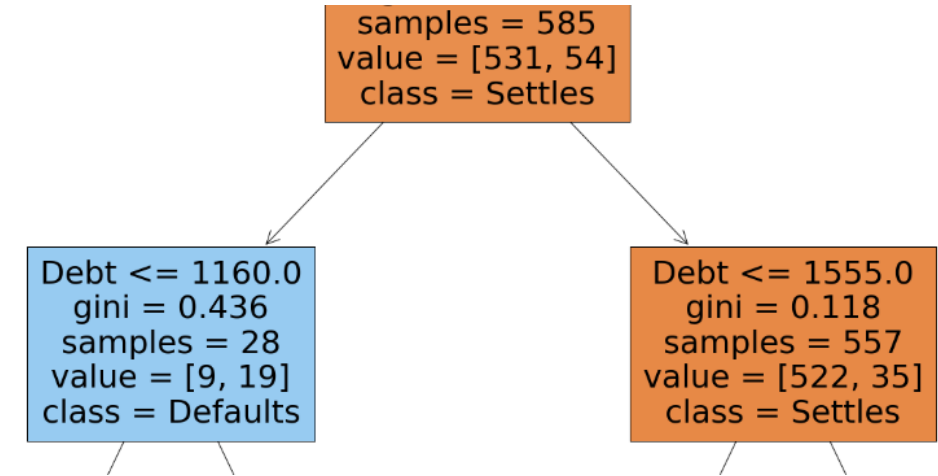
Calculate Gini Impurity for Potential Splits:

For each potential split of the data:

- Calculate the Gini impurity for each child node resulting from the split.
- Calculate the weighted average Gini impurity of the child nodes, where the weights are the proportions of samples that go into each child node.

Select the Split with the Lowest Weighted Gini Impurity:

- The split with the lowest weighted Gini impurity is selected because it indicates the highest purity for the child nodes.



The Weighted Average Gini for the split is:

$$\text{Weighted Gini} = \frac{28(0.436) + 557(0.118)}{585} = 0.1332$$



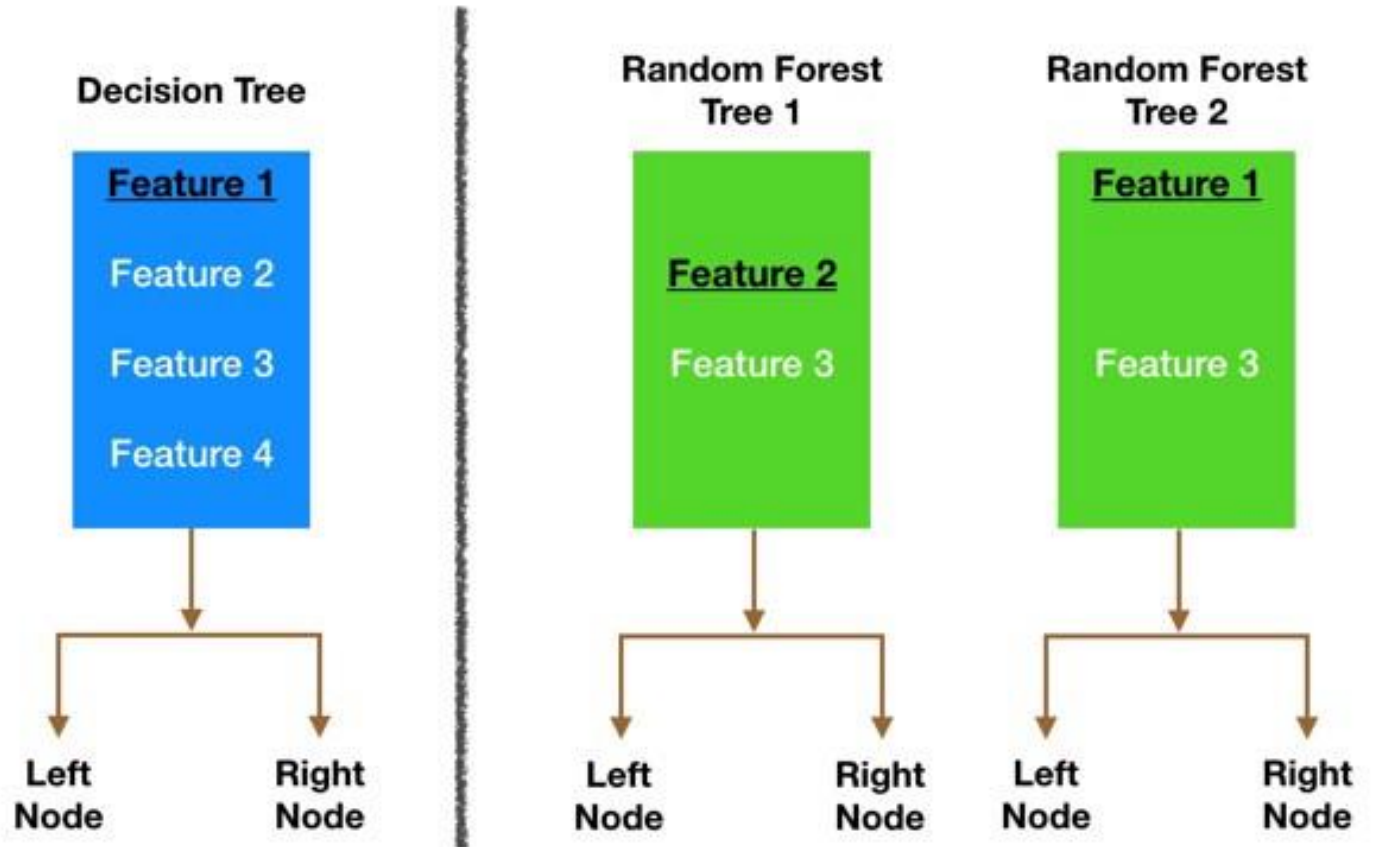
Extending Decision Trees

QA Random forests

A random forest takes some inherent disadvantages of decision trees and turns them into strengths.

It is an ensemble model built from a number of decision trees, hence its name.

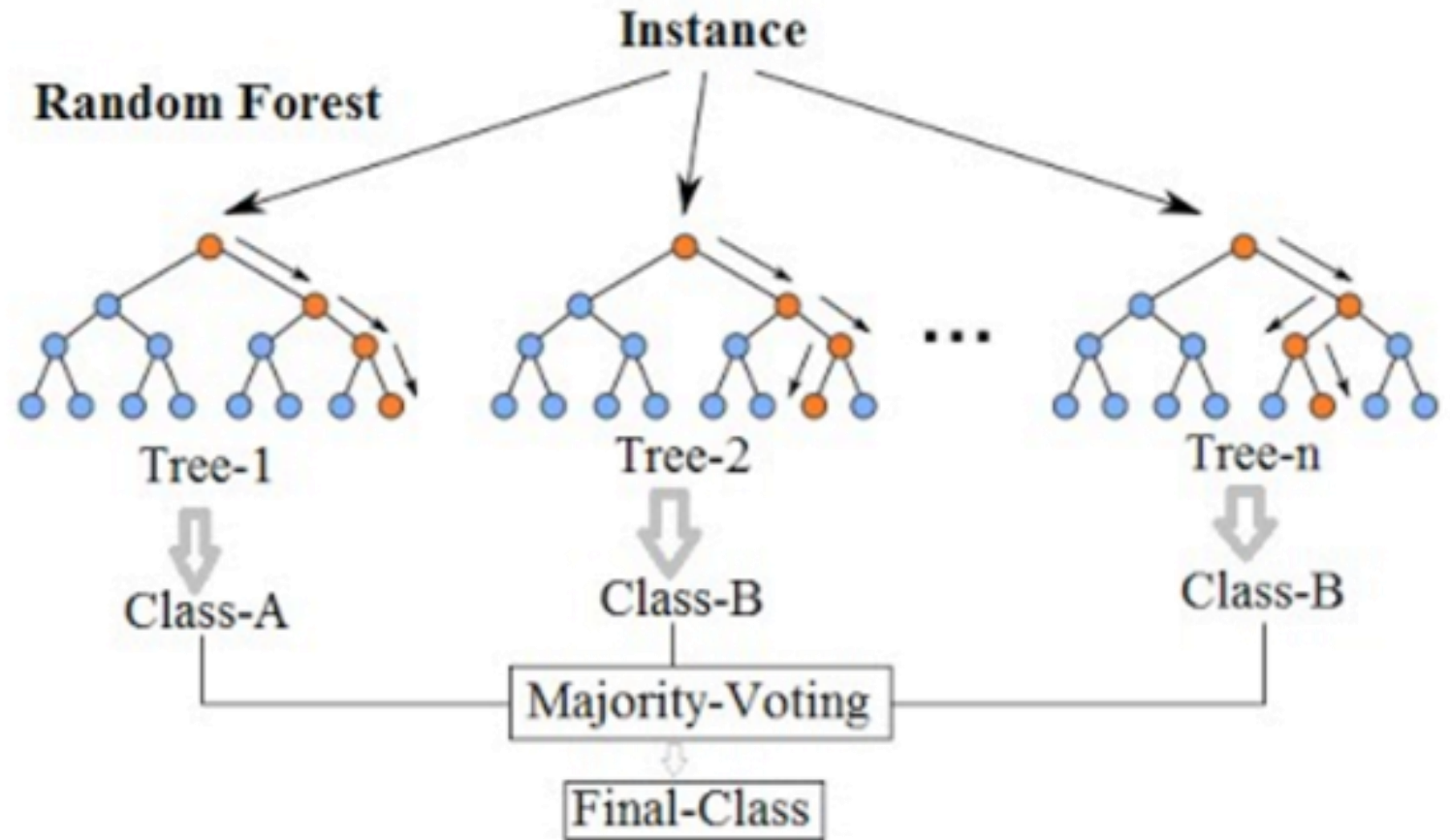
The fundamental concept behind a random forest is the ‘wisdom of crowds’; this is achievable by ensuring a diverse, or uncorrelated, forest of decision trees.



QA Random forests

Each tree in the random forest trains on a subset of the data, using bagging, with random features.

Then, when making a prediction, each tree votes, and the majority decision is the final class.





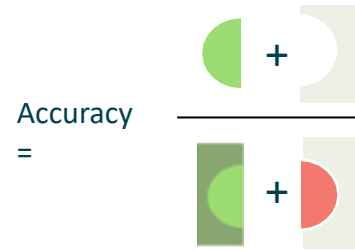
Classification Model Evaluation

- I. Metrics : Accuracy, Precision, Recall
- II. The Confusion Matrix

QA Model Evaluation Metrics

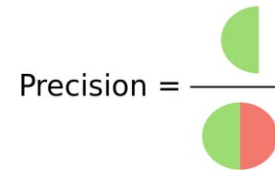
$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total}}$$

Accuracy: Overall accuracy, percentage of results that are True Positive and True Negative



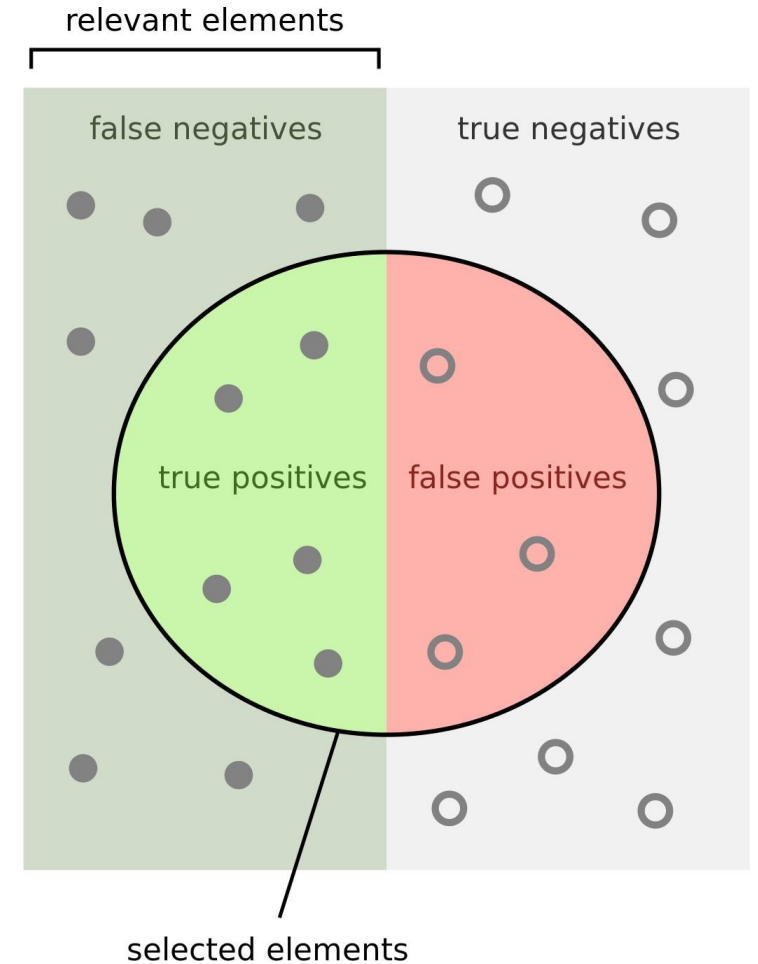
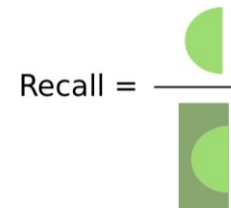
$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

Precision: The precision is intuitively the ability of the classifier not to label as positive a sample that is negative.



$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

Recall: Percentage of values that were correctly classified as True Positive across their relevant elements





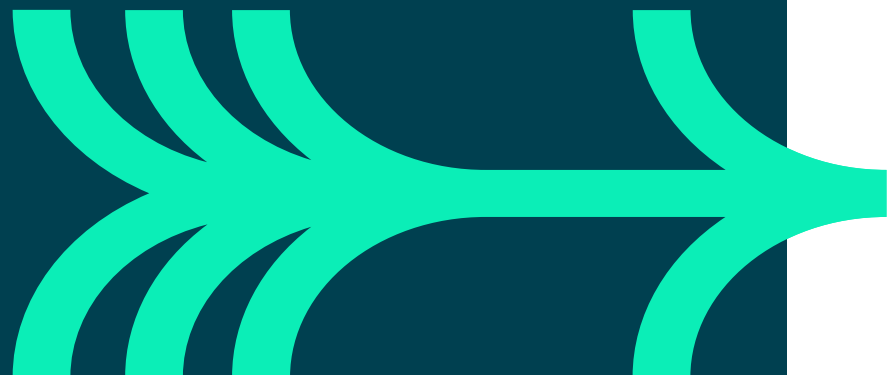
CLASSIFICATION METRICS: EVALUATION EXAMPLE

```
RandomForestClassifier(min_samples_leaf=10, n_estimators=30)
```

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

Settle	0.95	0.99	0.97	236
Default	0.73	0.38	0.50	21

accuracy			0.94	257
macro avg	0.84	0.68	0.73	257
weighted avg	0.93	0.94	0.93	257



QA Confusion Matrix

These metrics can be better understood and visualised by relating these 4 inputs in a grid format – **the Confusion Matrix**

- True Positive
- True Negative
- False Positive
- False Negative

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

QA Confusion Matrix

We have a classification model that classifies a picture that **contains a Hotdog as Positive**



QA Confusion Matrix

A model that classifies a picture that **contains a Hotdog as Positive**
Direct Message in the chat.

Where would you place the following:



Predicted Class

Positive
Negative

True Class

Positive

Negative

TP

FP

FN

TN

QA Confusion Matrix

Where would you place the following:



Predicted Class

Positive
Negative

True Class

Positive

Negative

TP

FP

FN

True negative



QA Confusion Matrix

Where would you place the following:



Predicted Class

Positive
Negative

True Class

Positive

Negative

TP

False positive



FN

True negative



QA Confusion Matrix

Lastly,



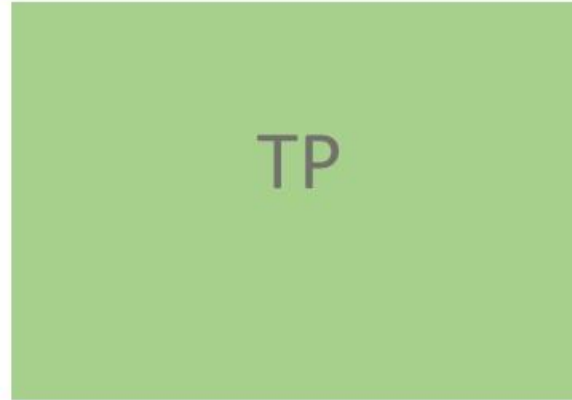
Predicted Class

Positive
Negative

True Class

Positive

Negative



False positive



False negative



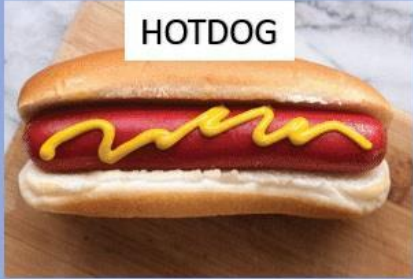
True negative



QA Confusion Matrix

Practical: Build a Confusion Matrix

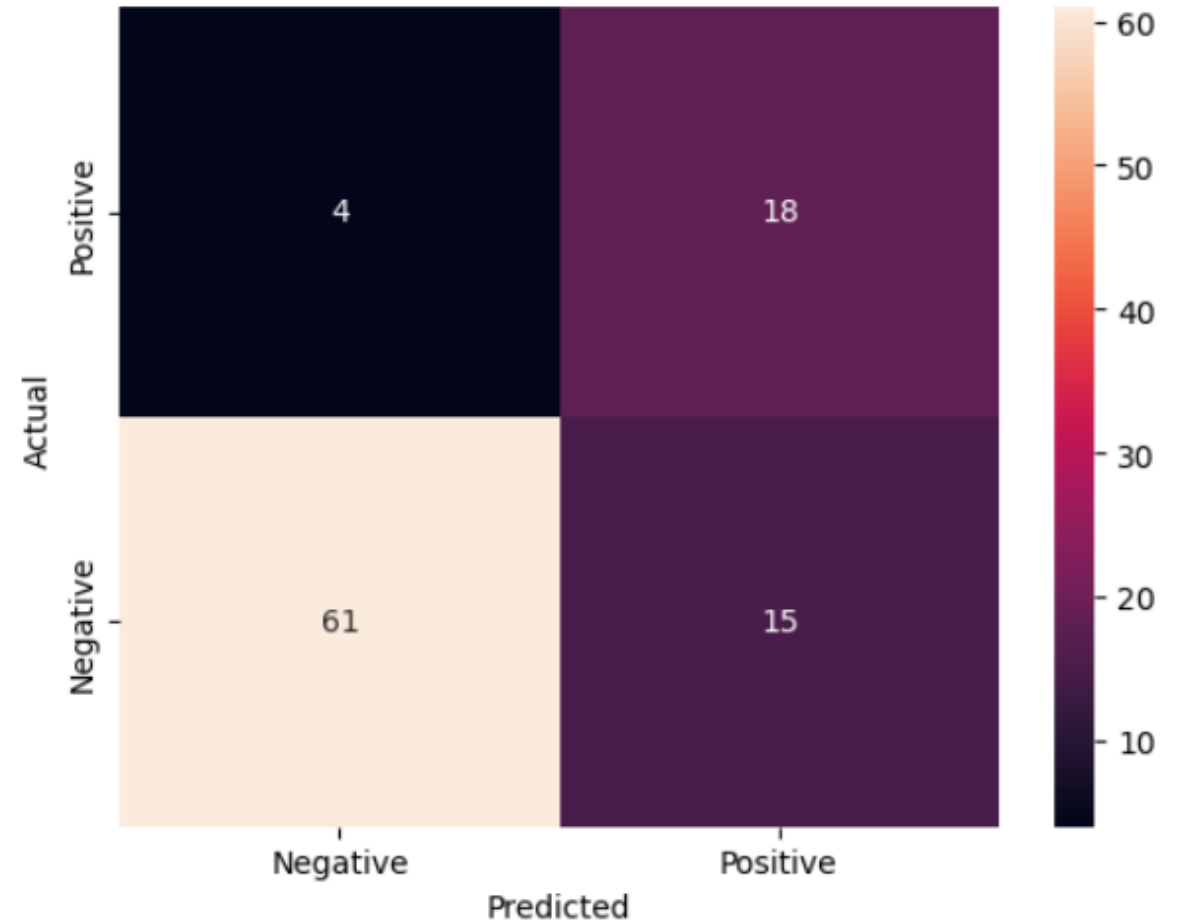
Be prepared to discuss its significance

		True Class	
		Positive	Negative
Predicted Class	Positive	<u>True positive</u> 	<u>False positive</u> 
	Negative	<u>False negative</u> 	<u>True negative</u> 

QA Confusion Matrix

A Confusion Matrix helps us:

- Deconstruct to understand the inputs for Evaluation Metrics
- Visualise all True and False classification counts in a presentable and understandable manner
- Provides feedback to tune our classification model thresholds to perform better operationally.



QA ROC – Receiver Operating Characteristics

Logistic regression is often used as a classifier to assign class labels to persons, items, transactions etc. It has the possibility of falsely classifying that a person would Churn.

ROC curve = True Positive Rate against False Positive Rate. ROC looks at every possible threshold value that results in a change of classification.

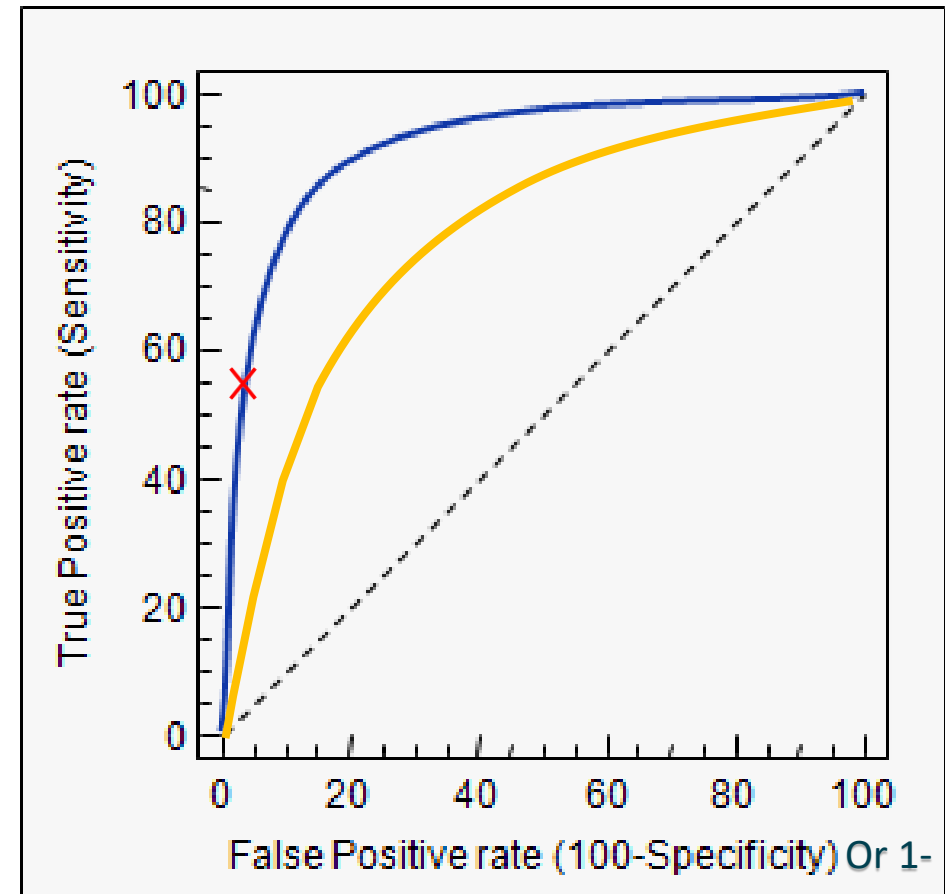
The Y axis is the Sensitivity or TPR.

The X-Axis is the 1 - Specificity or FPR.

The quicker the ROC curve approaches the top left of the graph the better the TPR, the more accurate the models predictions. (blue curve leads to model with more TPR and less FPR). The red cross is the 0.5 threshold, found by calculating the TPR and FPR for this model's 0.5 threshold.

$$\text{False Positive Rate (FPR)} = \frac{\# \text{ of false positives}}{\# \text{ of negatives}}$$

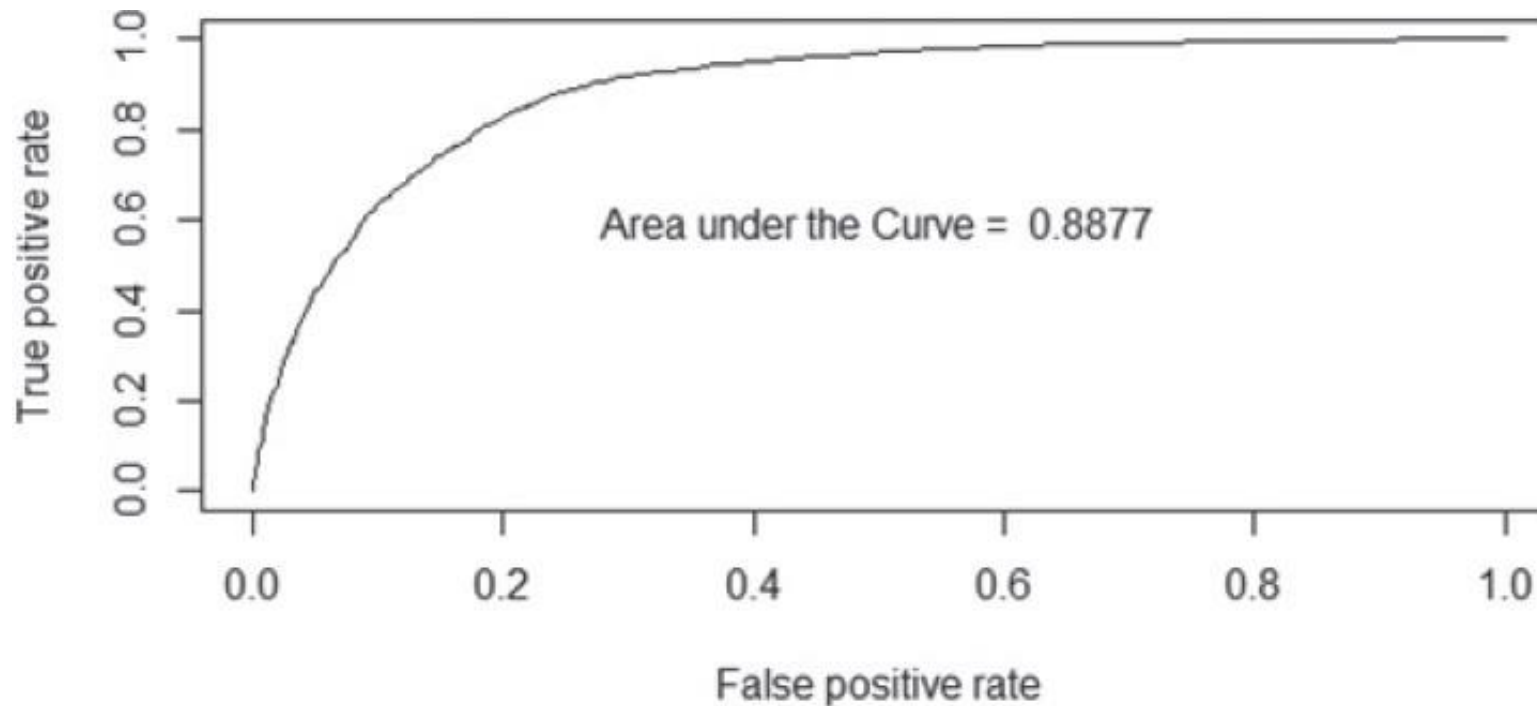
$$\text{True Positive : Rate (TPR)} = \frac{\# \text{ of true positives}}{\# \text{ of positives}}$$

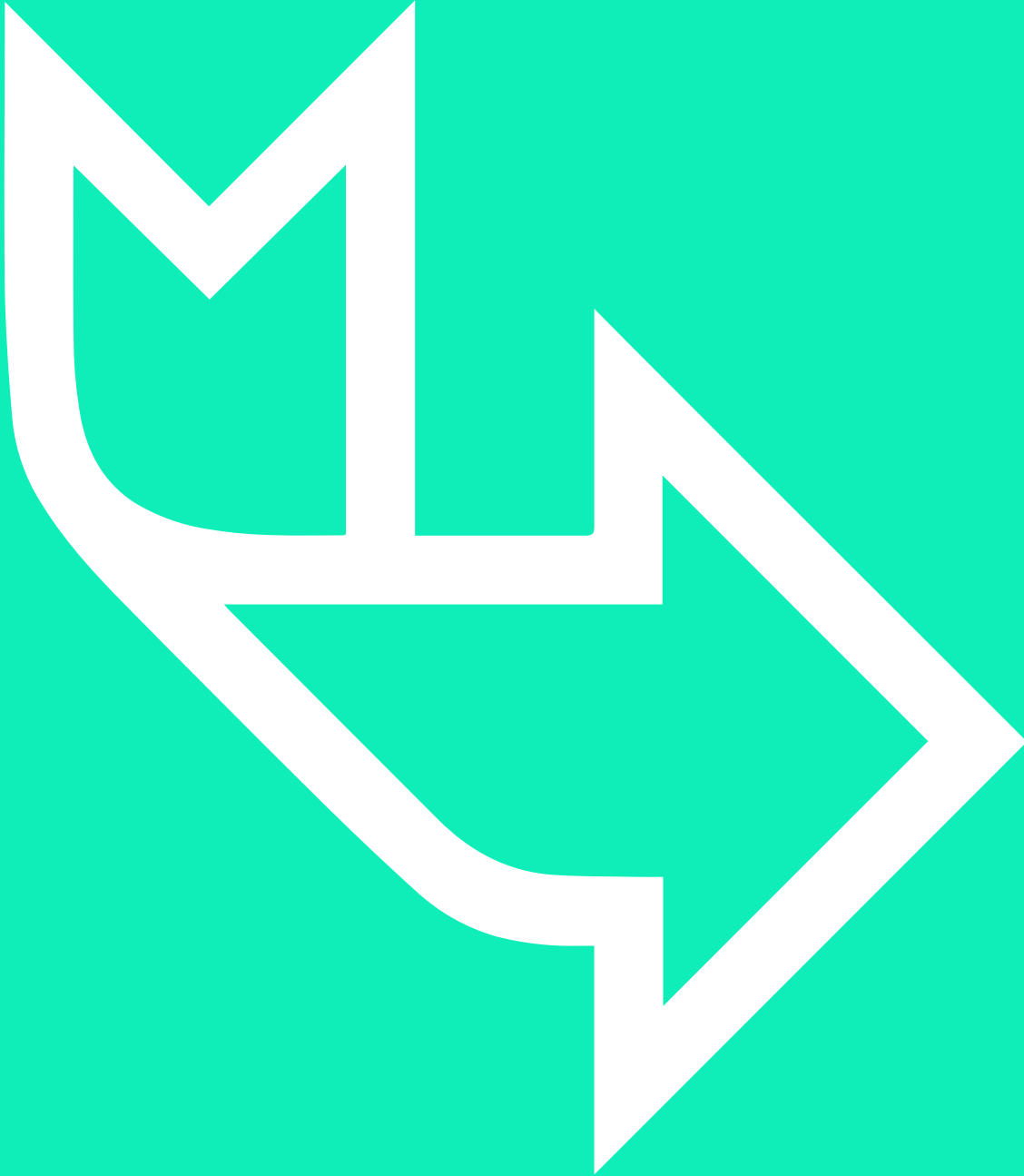


QA AUC – Area Under the Curve

AUC measures the model's discriminative ability, from 0-1. Its most useful as a comparative measure for several models.

An ideal model would have $AUC = 1$, able to discriminate between Churn and Remain with 100% accuracy. This means there is no overlap in y values between customers who remain and those who churn.





Assignment: Classification Modelling

Using the same prepared data, build a Decision Tree model and a Random Forest